



Short-term salinity prediction for coastal areas of the Vietnamese Mekong Delta using various machine learning algorithms: a case study in Soc Trang Province

Le Thi Thanh Dang^{1,2} · Hiroshi Ishidaira² · Ky Phung Nguyen³ · Kazuyoshi Souma² · Jun Magome²

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Abstract

Saltwater intrusion has significant and diverse impacts on agriculture, freshwater resources, and the well-being of coastal communities. To effectively address this issue, precise models for predicting saltwater intrusion must be developed, as well as timely information for reaction planning. In this study, a spectrum of machine learning (ML) methodologies, specifically Random Forest Regression (RFR), Support Vector Regression (SVR), Long Short-Term Memory (LSTM), Artificial Neural Network (ANN), Extreme Gradient Boosting (XGBoost), and Ridge Regression (RR), was systematically employed to predict salinity levels within the coastal environs of the Mekong Delta, Vietnam. The input dataset comprised hourly salinity measurements from Tran De, Long Phu, Dai Ngai, and Soc Trang stations and hourly water-level data from Tran De station and hourly discharge data from the Can Tho hydrological station. The dataset was partitioned into two distinct sets for the purpose of model development and evaluation, employing a division ratio of 75% for training (constituting 8469 observations) and 25% for testing (comprising 2822 observations). The results indicate that ML models are suitable for short-term salinity prediction, with a forecasting time of up to 16 h in this area. These research findings highlight the potential of machine learning in addressing saltwater intrusion and provide valuable insights for developing appropriate response policies. By leveraging the strengths of these models and considering the optimal forecasting time, policymakers can make informed decisions and implement effective measures to mitigate the impacts of saltwater intrusion in the Mekong Delta.

Keywords Salinity · Machine learning models · Optimal time · Coastal communities · Vietnamese Mekong Delta

Introduction

Water management is a critical concern for societies due to its impact on various aspects of life, affecting human health, agriculture, energy production, and environmental conservation (United Nations 2022; Belayneh et al. 2014). Saltwater intrusion is the phenomenon characterized by the infiltration of saline water with a typical salinity concentration of 4‰

into inland regions during events like high tides, elevated sea levels, or freshwater resource depletion. In recent years, there has been substantial global attention directed toward this problem due to the rise in sea levels resulting from climate change and global shifts (Nicholls and Cazenave 2010), which accelerate the encroachment of saltwater into freshwater bodies such as estuaries, deltas, and coastal aquifers.

Managing water and soil quality, especially in coastal areas and near estuaries, becomes a major challenge in the face of salinity intrusion (Tessler et al. 2015). The extent of this intrusion is contingent upon a multitude of factors (Liu et al. 2014), including tidal levels (Byung and Seung-Buhm 2013), wind patterns (Ralston et al. 2008a, 2008b; Uncles and Stephens 2011), water quality (Melesse et al. 2020), water discharge (Thanh et al. 2023; Lin and Qi 2017), water temperature (Pritchard 1955; Cameron and Pritchard 1963), and river channel slopes—all contributing to the extent and intensity of saltwater infiltration.

✉ Le Thi Thanh Dang
dtile@hcmus.edu.vn

¹ Faculty of Environment, University of Science, Vietnam National University Ho Chi Minh City, 227 Nguyen Van Cu Street, District 5, Ho Chi Minh City, Vietnam

² Interdisciplinary Centre for River Basin Environment, University of Yamanashi, 4-3-11 Takeda, Kofu, Yamanashi 400-8511, Japan

³ Thu Duc People's Committee, 168 Truong Van Bang Street, Thanh My Loi, Thu Duc City, Ho Chi Minh City, Vietnam

Saltwater intrusion significantly impacts ecosystems and water utilization (Singh et al. 2020). High salt concentrations disrupt ecological processes, leading to adverse effects on biodiversity and productivity in aquatic and terrestrial environments (Pereira et al. 2019). Agricultural systems are particularly affected by salinity intrusion (Machado and Ricardo 2017; Yang et al. 2022; Sayidjakhon et al. 2023). Soil salt accumulation hinders crop growth, leading to reduced yields and economic losses for farmers. Elevated salinity levels in irrigation water also harm plant health and reduce fertilizer effectiveness, exacerbating agricultural challenges (Marcelis et al. 2000). Addressing salinity intrusion necessitates a multidisciplinary approach, encompassing scientific research, monitoring, and effective management strategies. Sustainable water management practices, like saltwater intrusion barriers, efficient irrigation techniques, and water conservation, are vital. Understanding the long-term effects of climate change and sea-level rise on salinity intrusion is essential for developing adaptation and mitigation strategies (Somboonsuke et al. 2018; Wawargita and Rahmat 2016; Tran et al. 2021).

Salinity intrusion poses a major challenge for the Vietnamese Mekong Delta (Ketabchi et al. 2016; Le et al. 2023). The influence of salinity intrusion on the Vietnamese Mekong Delta is substantial, with negative consequences for agriculture (Dang and Nguyen 2019; Dang et al. 2023). Additionally, the quality and quantity of freshwater resources, crucial for drinking, irrigation, and aquaculture, are affected (Tran et al. 2021; Tho et al. 2008). Salinity intrusion also negatively impacts the health and well-being of local communities, who rely on freshwater sources for their daily activities (Nguyen et al. 2020; Tuan 2010; Dung et al. 2022).

Numerous studies have been conducted on saltwater intrusion, categorizing approaches into steady-state and unsteady-state models (Gou et al. 2019; Robinson et al. 2018). Numerous research endeavors have utilized these numerical models to examine saline intrusion and its consequences. For example, in a study conducted by Mittelstel et al. (2015), they employed the SWAT model to examine saltwater intrusion within the North Fork River basin in the USA, while Maleki et al. (2021) used the same model to simulate natural saline processes in watershed systems. Hadout et al. (2015) applied the HEC-RAS model to assess the dispersion of salinity within Morocco's Sebou estuary, while Saran et al. (2018) utilized HEC-RAS to model the patterns of salinity in the Rupsha-Passur river system. Etemad-Shahidi et al. (2008) used the MIKE 11 model for Danshuei estuary hydrodynamics. Quang et al. (2014) assessed Mekong estuary saline intrusion under dredging effects with MIKE 11. Nguyen et al. (2016) simulated Mekong Delta saline intrusion's response to sea-level rise from 2015 to 2100 using MIKE 11. Kashefi et al. (2010) predicted the salinity

of the Karun River based on hydrological parameters using the MIKE 11 model, while Keilholz et al. (2015) assessed underground flow in the Tarim Basin in Germany using the MIKE SHE model. In general, numerical models offer accurate calculation results and comprehensive descriptions of physical phenomena. However, their application in studying saltwater intrusion is not without challenges. The use of multi-dimensional numerical models is constrained due to the scarcity of accessible observed chlorinity and hydrological data, along with the absence of comprehensive details concerning extensive estuarine topography. These factors, as emphasized by Xu et al. (2019a, 2019b), are essential for accurately representing and analyzing the dynamics of saltwater intrusion in estuaries. Therefore, addressing these limitations becomes crucial for enhancing the effectiveness of numerical models in studying saltwater intrusion.

Big data and artificial intelligence provide innovative solutions to longstanding problems. In recent years, machine learning and deep learning techniques have significantly advanced modeling and prediction of hydrological, hydrodynamic processes, and climate change, offering advantages including enhanced improved accuracy, computation performance (Ardabili et al. 2019). In the fields of hydrology, salinity intrusion, and earth systems, research increasingly applies machine learning algorithms to predict water resource variations. These methods serve various purposes, including flood stage prediction (Pao-Shan et al. 2006), daily atmospheric pollen concentration estimation (Zewdie et al. 2019), discharge prediction (Kovacevic et al. 2018), time-series flooding prediction (Hu et al. 2019), and forecasting drought-related factors such as SPIE (Shen et al. 2019), salinity prediction (Liu et al. 2014; Dong et al. 2010; Rohmer and Brisset 2017). Machine learning techniques, particularly artificial intelligence-based models, are being increasingly recognized for their ability to model complex, nonlinear relationships within hydrological and environmental datasets. A relevant example can be found in the work of Alizadeh et al. (2018), who conducted a comparative assessment of the performance of Artificial Neural Networks (ANN), Support Vector Regression (SVR), and Extreme Learning Machine (ELM) for the prediction of water quality in Hilo Bay, Hawaii. Their findings revealed that both ANN and SVR exhibited superior capabilities in handling complex, nonlinear, and high-dimensional datasets, typical of environmental data. However, the models' predictive performance was found to be heavily influenced by the choice of input variables, particularly streamflow, which was identified as a critical determinant in improving forecast accuracy. Similarly, Rath et al. (2017) employed a Bayesian model, an advanced variant of ANN, to forecast salinity levels in San Francisco Bay. The Bayesian model outperformed traditional ANN models, demonstrating significantly higher accuracy. Notably, their analysis suggested

that further improvements in model performance could be achieved by increasing the training dataset size. In a comparable study, Yoon et al. (2017) compared the performance of Support Vector Machine (SVM) and ANN for predicting salinity dynamics between freshwater and saline environments. Their results indicated that SVM exhibited superior stability across a range of scenarios, whereas ANN delivered higher accuracy in specific instances. Khan et al. (2021) utilized Random Forest Regression (RFR) to model surface water salinity, underscoring the method's effectiveness in reducing model complexity and enhancing its applicability for water resource management.

In this study, a range of machine learning methodologies was employed, each selected based on its specific advantages for handling the intricacies of hydrological data. Random Forest Regression (RFR) is well-regarded for its ability to manage high-dimensional and complex datasets, mitigate overfitting, and provide robust predictions. Support Vector Regression (SVR) is particularly effective for both linear and nonlinear data, ensuring high precision in various predictive contexts. Long Short-Term Memory (LSTM), a sophisticated recurrent neural network architecture, excels in capturing temporal dependencies, rendering it highly suitable for time-series forecasting applications. Artificial Neural Networks (ANN) provide flexibility in modeling nonlinear relationships and are capable of detecting complex, intricate patterns in the data, which is vital for modeling multifaceted hydrological processes. Extreme Gradient Boosting (XGBoost) is a powerful ensemble learning algorithm that optimizes predictive performance through a combination of weak learners, reducing overfitting while enhancing overall accuracy. Finally, Ridge Regression (RR), a regularization technique, enhances model stability and accuracy by addressing multicollinearity in the data, further improving its robustness and reliability for hydrological applications. These methodologies were employed to assess and compare their efficacy in hydrological modeling, with the goal of identifying the most suitable model for the research objectives.

There is no denying that salinity intrusion is considered a remarkable threat for Vietnam Mekong Delta. Managing this challenge in the delta requires accurate simulation to effectively utilize water resources. Both physical models and machine learning algorithms have their strengths and weaknesses when it comes to simulating saltwater intrusion in this region. Physical models offer a more in-depth comprehension of the underlying processes but can be computationally demanding and require extensive input data. In contrast, machine learning algorithms demonstrate exceptional proficiency in real-time applications and adeptly manage vast and complex datasets. The decision to employ either physical modeling or machine learning hinges on the particular goals of the application and the accessibility of data and resources.

Previous studies on salinity prediction in the Vietnamese Mekong Delta have employed certain algorithms, but they have typically focused on specific areas (Tran et al. 2021) or relied on a single algorithm to predict saltwater intrusion (Tran et al. 2022). This study seeks to offer a comprehensive insight into the potential of machine learning for salinity prediction by considering various input parameters that affect saltwater intrusion. By incorporating different input parameters such as hourly salinity data, discharge, and water-level data, this study is fundamentally centered on the development and investigation of algorithms for short-term salinity prediction within the Vietnamese Mekong Delta, offering new insights into their applicability. Two specific aims have been defined:

- (1) Develop machine learning algorithms tailored for short-term salinity prediction in VMD.
- (2) Compare the effectiveness of these different algorithms in predicting salinity, specifically in VMD. Model performance will be assessed using statistical indices and graphical comparisons.

Study area

This research was conducted in Soc Trang Province, located in Vietnam's Mekong Delta region, between latitudes 9°2'–9°56' N and longitudes 105°33'–106°33' E. Soc Trang shares borders with Hau Giang, Bac Lieu, Tra Vinh, and the East Sea. Covering an area of 3311.87 km² and home to 1,206,819 people, the province is characterized by a dense canal system influenced by semi-diurnal tides, with fluctuations ranging from 0.4 to 1 m. This area experiences a tropical monsoon climate with two distinct seasons: a rainy season (May–October) dominated by the southwest monsoon and a dry season (November–April) influenced by the northeast monsoon. Annual rainfall averages 1660–2230 mm, with 90% concentrated in the rainy season. The province receives significant solar radiation (140–150 kcal/cm²) and 2545 h of sunshine annually. Soc Trang benefits from its unique geographical location, as it is situated at the point where the Hau River flows into the Southeast Sea. This favorable terrain provides conducive conditions for the integrated development of the marine economy in the province. Additionally, Soc Trang has a coastline of 72 km, making it susceptible to significant damage from annual saline intrusion, a common issue in the Mekong Delta.

According to statistics from Soc Trang Province's Sub-department of Irrigation and Flood and Storm Control, in March 2020, the highest recorded salinity levels were observed at various stations. For example, at Tran De station on the Hau River, salinity reached 22.5‰, while at Dai Ngai, located 35 km from the Hau River, salinity was

measured at 10.5‰, and at An Lac Tay, it reached 5.5‰. Salinity affects almost every part of Soc Trang Province, and the March wave of saltwater intrusion was more severe than that observed in mid-February 2020, penetrating 30–40 km further inland. This situation has had a significant impact on both the production and daily lives of the people in the province (Le et al. 2023; Dung et al. 2021; Soc Trang Statistical yearbook 2021; Soc Trang Statistical yearbook 2020).

While many studies have examined the phenomenon of saltwater intrusion in the Mekong Delta using various methods such as modeling, remote sensing, and rapid assessment, most of these studies have focused on severely affected provinces like Tien Giang, Can Tho, and Ben Tre or have addressed the entire Mekong Delta region as a whole. In contrast, provinces like Soc Trang, which face

similar challenges, have received comparatively less attention in the field of this research (Fig. 1).

Material and methodology

General approach

This research involves the application of Random Forest Regression (RFR), Support Vector Regression (SVR), Ridge Regression (RR), Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and Extreme Gradient Boosting (XGBoost) for predicting salinity in the region. These techniques were selected based on their potential to effectively capture and model the complex relationships between salinity and various influencing factors. In order to achieve

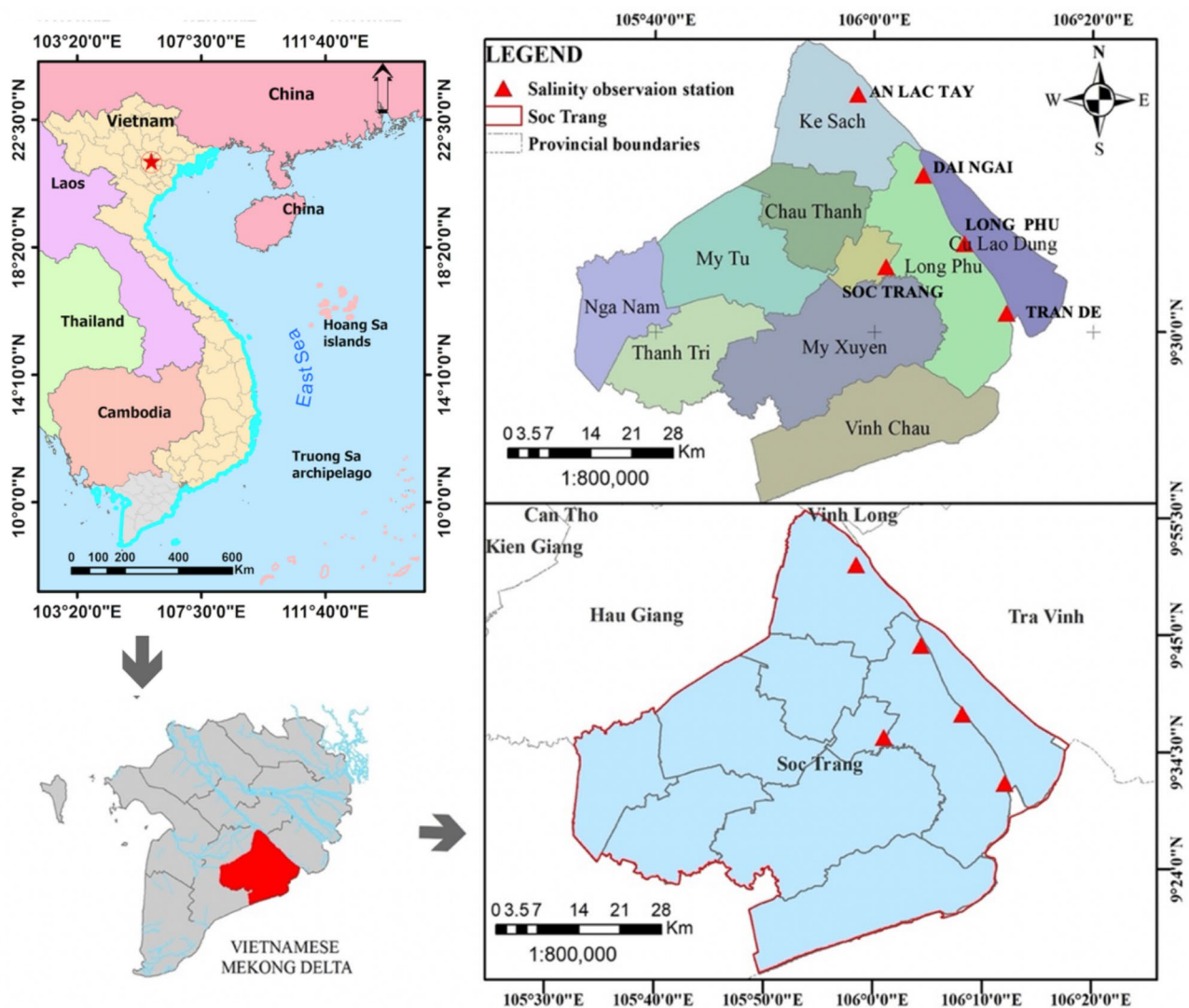


Fig. 1 Study areas

this objective, several tasks need to be undertaken. The diagram presented below illustrated the research framework employed in this study (Fig. 2).

Overall, the research workflow involves several key tasks, including data collection and processing, training set—testing set split, model training and testing, and evaluation. During the data pre-processing phase, null values, extreme outliers, and repeated values are removed using a Python library, resulting in a refined dataset. Exploratory Data Analysis (EDA) is then conducted on the final dataset to analyze the impact of different parameters on salinity at Tran De. To assess the models' performance, the dataset is split into training dataset and testing dataset. RF, XGBoost, RR, SVR, ANN, and LSTM models are applied to predict salinity. The models are evaluated using metrics such as R^2 , RMSE, MSE, MAE. We employ Grid Search CV function from the sklearn library to optimize the models' hyperparameters and select the best configuration (Bergstra and Bengio 2012). This function systematically explores defined hyperparameters to find the optimal parameters for the model fitting process.

Selection input parameter

In order to predict salinity, we have utilized six machine learning algorithms: SVR, ANN, RFR, XGBoost, RR, and LSTM. We chose the input variables through a comprehensive literature review and our own expertise. We have referenced previous studies and considered the specific characteristics of the study area to determine the appropriate input parameters. Previous research has highlighted the significance of discharge for salinity, as well as water level on the distribution and density of salinity concentrations

(Haddout and Maslouhi 2018; Lin et al. 2019; He et al. 2020; Zhou et al. 2020; Nicholas and William 2022; Changklom et al. 2022; Thanh et al. 2023). Additionally, studies have indicated that current salinity is highly dependent on past salinity values and the salinity of neighboring stations (Yoon et al. 2017; Lin et al. 2019; He et al. 2020; Changklom et al. 2022). Therefore, for this study, we considered discharge, water level, and salinity as prerequisite variables for predicting salinity. The selection indicators for these variables prioritized sufficiency, relevance, usability, and data accessibility. A detailed list of the input parameters for salinity prediction can be found in Table 1.

Data

The dataset for the investigation comprises hourly data collected during the dry season, spanning from January to May, for the years 2013 to 2020. However, based on measured data and the Soc Trang Statistical Yearbook (2016, 2017, 2018, 2019, 2020), saltwater intrusion in the area typically begins in February, peaks around March to April, and gradually subsides by the end of May. Therefore, the period from February to May was selected based on the characteristics of saltwater intrusion in the study area. The field data consisted of salinity measurements taken every two hours at the Tran De, Long Phu, Dai Ngai, An Lac Tay, and Soc Trang stations, along with hourly water-level measurements from the Tran De gauge and hourly discharge data from the Can Tho hydrological station. For this study, to maintain consistency with the model's temporal resolution and facilitate the analysis of hydrological events, all data were standardized to a two-hour time step. For the training and testing sets, we used

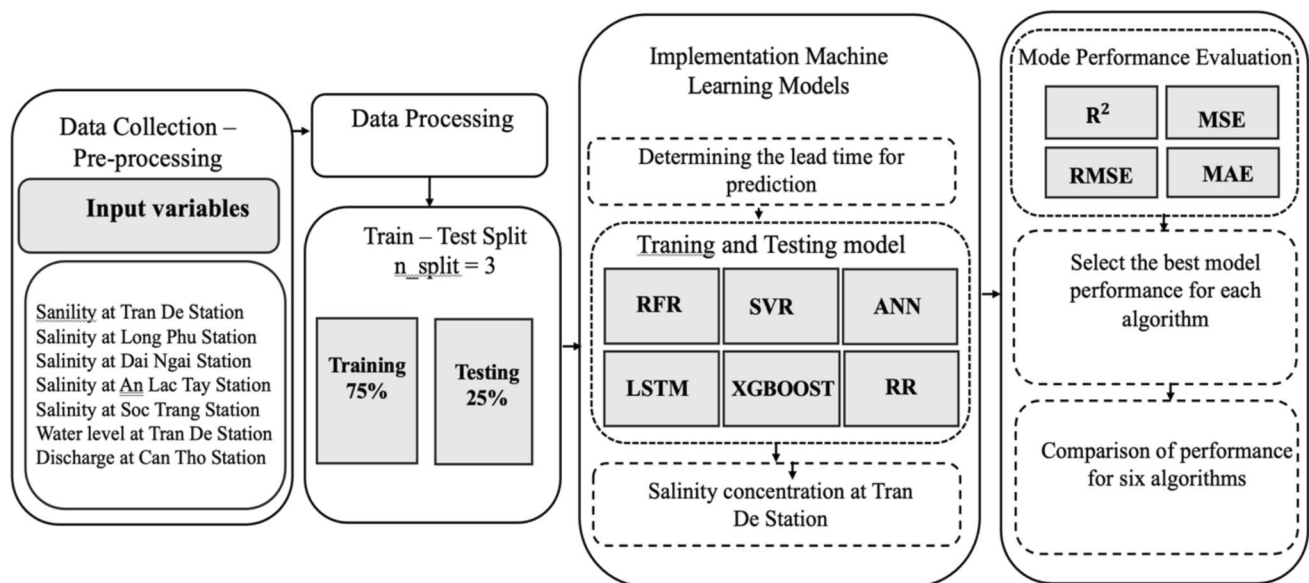


Fig. 2 Research work flow diagram of this study

$n_split=3$ to divide the data. The first 8469 observations (75%) were designated as the training set, while the final 2822 observations (25%) were allocated for the testing set. The description of the dataset details and the methodology employed in this investigation are shown below (Table 2).

The attributes of the gathered data are shown in Table 3. The statistics indicate that the data values exhibit fluctuations across different intervals. In particular, water discharge and water-level data have a wider range compared to the other parameters. Additionally, the dispersion intervals of the data variables show significant variations.

The distribution and relationship between input variables are illustrated in Fig. 3 using the seaborn module in Python.

Figure 3 displays pairwise scatter plots with kernel density estimation (KDE). This pairplot provides a clear visualization of the relationships between variables in the study area. It reveals that salinity concentrations at the stations in the study area exhibit a strong correlation, indicating a close relationship between them. On the other hand, variables like discharge and water level show relatively low correlations

with salinity concentration, especially in the case of system values. Specifically, the correlation coefficient between water level at the Tran De station and the salinity concentration at the same station is 0.25, while the correlation coefficient between discharge at the Can Tho station and the salinity concentration at the Tran De station is 0.39.

Figure 4 focuses on the influence of each parameter on others. Diagonal correlation heatmaps, created using the seaborn module in Python, display the correlation coefficients (R) of each parameter on others as well as multiple inputs with salinity at Tran De station. As we can see in Fig. 4, there are relatively moderate correlations between parameters overall. The value of correlation coefficients (R) is between -1 and $+1$, where $r=1$ represents a perfect positive linear correlation; $(0.8 < r < 1)$ indicates a strong positive correlation, $(0.4 < r < 0.8)$ represents a moderate positive correlation, $(0 < r < 0.4)$ indicates a weak positive correlation, $(r=0)$ means no correlation, $(-0.8 < r < -0.4)$ indicates a weak negative correlation, $(-1 < r < -0.8)$ represents a moderate negative correlation.

Table 1 Input parameters for study area

No	Input variables	Source
1	Discharge	Adapted from Liu et al. (2014), Lin et al. (2019), Zhou et al. (2020), He et al. (2020), Zhou et al. (2021), Thanh et al. (2023)
2	Water level	Adapted from Haddout and Maslouhi (2018), Lin et al. (2019), He et al. (2020), Zhou et al. (2020), Nicholas and William (2022), Changklom et al. (2022), Thanh et al. (2023)
3	Salinity	Adapted from Sun and Koch. (2001), Dong et al. (2010), Qiu and Wan (2013), Yoon et al. (2017), Lin et al. (2019), He et al. (2020), Changklom et al. (2022) $m^3 s^{-1}$

Table 2 Gathering data for predict salinity

Stations	Indicators	Time step	Time series
Tran De	Salinity	Two hours	2013/02/11/01:00–2013/05/28/23:00
Long Phu			2014/02/01/01:00–2014/05/31/23:00
Dai Ngai			2015/02/02/01:00–2015/05/31/23:00
An Lac Tay			2016/02/01/01:00–2016/05/31/23:00
Soc Trang			2017/02/01/01:00–2017/05/31/23:00
Can Tho	Discharge		2018/02/02/01:00–2018/05/02/23:00
Tran De			2019/02/01/01:00–2019/05/31/23:00
	Water level		2020/02/01/01:00–2020/05/31/23:00

Table 3 Descriptive statistics input variables

	Tran De	Long Phu	Dai Ngai	An Lac Tay	Soc Trang	Water level	Discharge
Count	11,291	11,291	11,291	11,291	11,291	11,291	11,291
Mean	10.79	6.24	2.14	0.37	1.95	35.73	1735.86
Std	5.15	4.34	2.47	1.06	1.59	87.92	9820.78
Min	0.2	0	−0.02	−0.01	0	−185	−20,300
First quartile (25%)	6.80	2.9	0.30	0	0.7	−35	−7000
Median (50%)	10.47	5.32	1.2	0	1.60	48	3140
Third quartile (75%)	14.3	8.91	3.16	0.2	2.8	108	11,100
Max	27.40	23.10	14.20	8	9	225	19,000

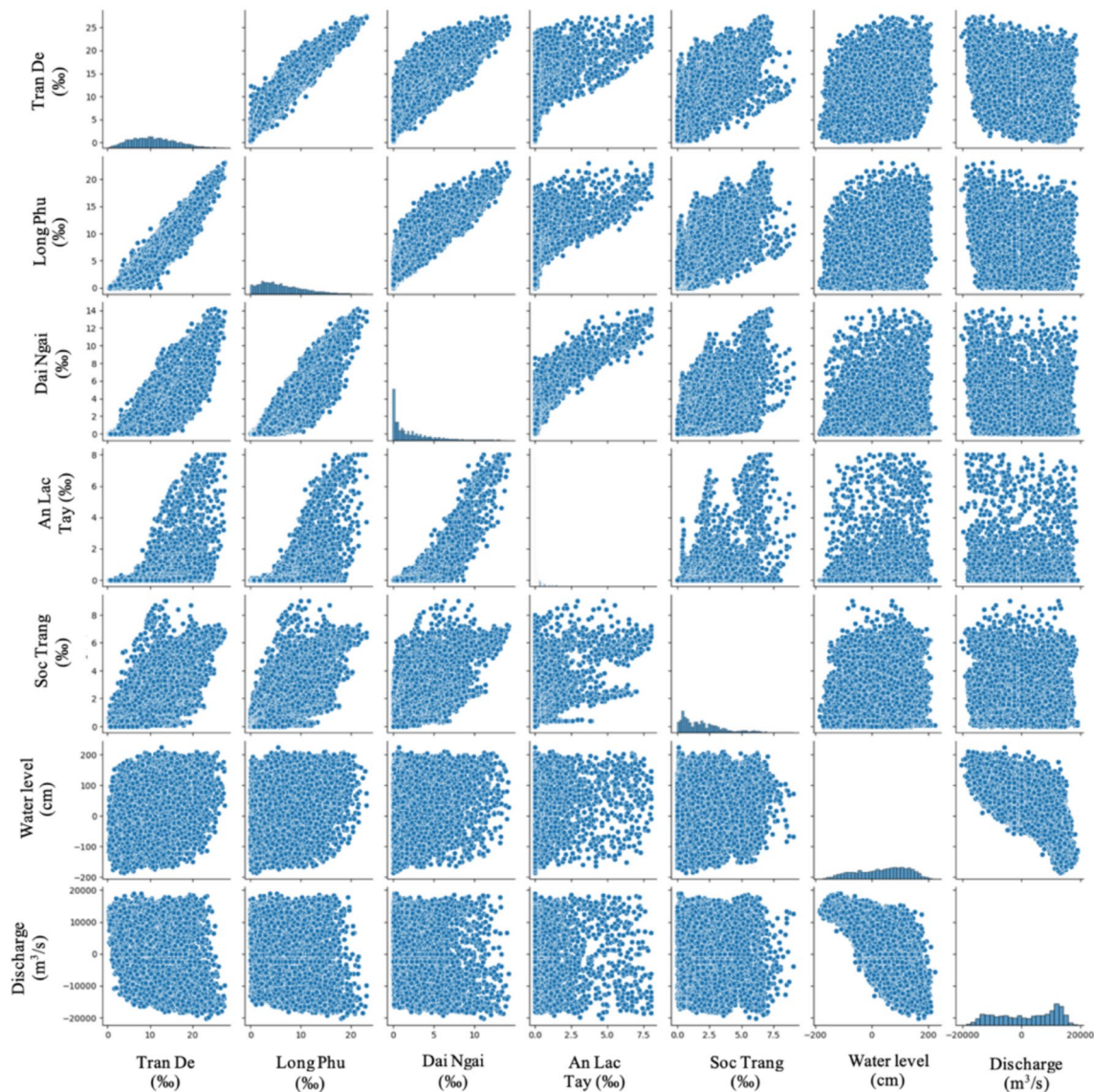


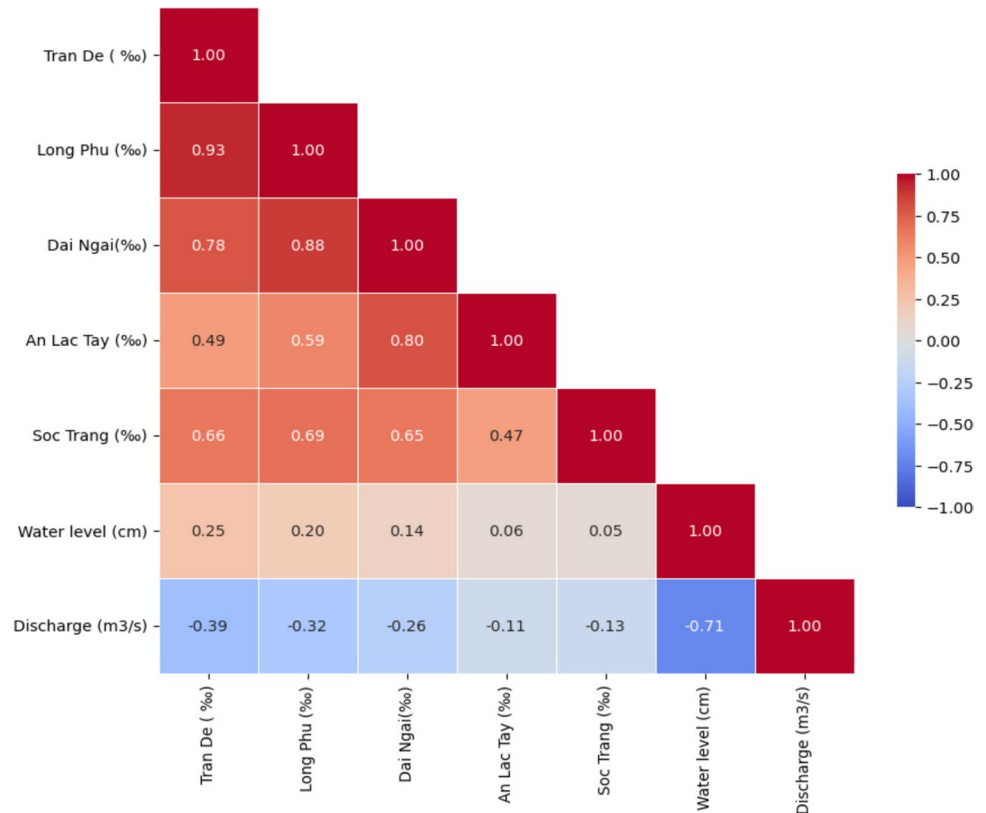
Fig. 3 Data distribution and their correlation

strong negative correlation, and ($r = -1$) represents a perfect negative linear correlation (Zar 1999).

Data normalization

As indicated in Table 1, the presence of numerical variables with different units and scales can exert a notable influence on the model's effectiveness. These variations in units and scales may lead to certain features exerting a disproportionate influence on the model. To mitigate this issue, it is crucial to normalize the range of all features. By normalizing the features, each feature contributes proportionately to the model, ensuring a fair and unbiased representation of the data.

Data normalization aims to ensure internal consistency within the data by standardizing the content and format of each data type. Within the scope of this study, the collected data comprise attributes with varying scales and units, resulting in wide variations in the original data values. To alleviate the influence of such heterogeneous data on the application of machine learning algorithms, the study proceeds to adjust the attribute ratios so that all attributes have a consistent scale. This is achieved through data normalization, which ensures that each data type possesses the same content and format. The chosen normalization method in this study is MinMaxScaler, with the following formula:

Fig. 4 Heat map of the correlation matrix

$$x'_i = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)} \quad (1)$$

where x'_i is the normalized value; x_i is the original value; $\min(x_i)$ is the minimum value of the variable x ; $\max(x_i)$ is the maximum value of the variable x ;

Machine learning models

Machine learning is widely recognized as a valuable tool for improving decision-making in predicting the saline intrusion. Numerous studies have highlighted its importance in providing comprehensive guidance and valuable insights, thereby assisting farmers in mitigating farming losses. In this study, several machine learning models (MLMs) were examined, including Support Vector Regression (SVR), Random Forest Regression (RFR), Extreme Gradient Boosting (XGB), Long Short-Term Memory (LSTM), Artificial Neural Networks (ANN), and Ridge Regression (RR). The selection of these particular methods was driven by the size of the dataset and the specific characteristics of the prediction task.

Support Vector Regression (SVR)

SVR is a machine learning technique designed for regression tasks, aiming to predict continuous output variables. It

is derived from Support Vector Machines (SVM) and was first introduced by Drucker et al. (1997) based on Vapnik's support vectors concept (Cortes and Vapnik 1995).

An essential aspect of SVR involves employing kernel functions, facilitating the capture of nonlinear associations between input variables and the target variable. These kernels perform inner product calculations within the transformed feature space, enabling SVR to work seamlessly in a higher-dimensional context without requiring an explicit transformation. SVR can handle both linear and nonlinear regression problems by utilizing different types of kernels. The selection of the appropriate kernel depends on data characteristics and the preferred level of flexibility in the regression model. SVR has applications in various domains, including finance and economics (Jaramillo et al. 2017), engineering (Jindal et al 2020), and environmental sciences (Chang and Lin 2011).

Random Forest Regression (RFR)

RF is one of the prominent supervised machine algorithms that are effective for both categorical and regression tasks. It was first introduced by Breiman (2001). It employs an ensemble of decision trees to improve prediction accuracy.

RFR is a variant of the Random Forest algorithm, which combines predictions from multiple machine learning models to achieve higher accuracy. In Random Forest

Regression, decision tree ensembles are built from random training data subsets, with splits chosen based on criteria like mean squared error or mean absolute error. Predictions from all trees are averaged during the prediction phase to enhance accuracy and reduce variance (Fig. 5).

Random Forest Regression has various applications across different domains due to its versatility and robustness, including agricultural (Filippi et al. 2019; Xu et al. 2022), environmental science (Nicolas and William 2022), energy (Zhou et al. 2016; Aditya et al. 2020).

Long short-term memory (LSTM)

LSTM is a specialized form of recurrent neural network (RNN) architecture widely used for analyzing and predicting sequential data. This innovative architecture was first introduced by Hochreiter and Schmidhuber (1997).

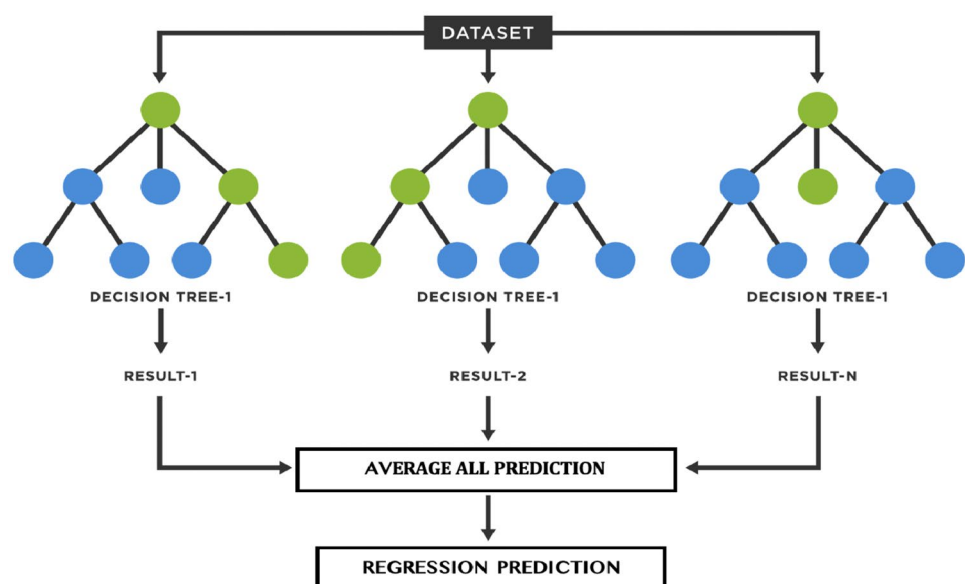
The LSTM architecture consists of memory cells, which are the fundamental building blocks of the network. Each memory cell, which includes the forget gate, the input gate, and the output gate. The forget gate in an LSTM plays a crucial role in determining which information from the previous cell state should be retained and which should be forgotten. The input gate serves as the factor that dictates the extent to which fresh data should be integrated into the memory cell. It controls the flow of new data into the cell, allowing the network to decide which parts of the input are important and should be stored. It consists of three main components: (1) sigmoid layer (input gate layer); (2) tanh activation function layer, which ensures that values are in the range $[-1, 1]$, allowing the network to capture both positive and negative information; (3) by combining the outputs of the sigmoid layer and the tanh

layer, the input gate determines the new information to integrate into the cell state. Lastly, the output gate determines how much data from the memory cell contribute to shaping the LSTM's output. This gate regulates the transfer of information from the cell to the output layer, allowing the network to carefully select and produce the relevant information. It includes both a sigmoid layer and a tanh layer.

Extreme Gradient Boosting (XGBoost)

XGBoost, an influential tree-based ensemble algorithm created by Chen and Guestrin (2016), gained widespread use in data science due to its potent performance and versatility. It builds on Friedman's gradient boosting framework from 2001 by combining multiple weak predictive models, such as decision trees, and iteratively enhancing their predictions through gradient descent. XGBoost's strength lies in its capacity to handle complex datasets and capture intricate variable relationships. It achieves this by using various functions to estimate the final prediction, which is a weighted sum of predictions from individual trees. The weights assigned to each tree depend on their performance and contribution to minimizing the loss function. By optimizing the loss function, XGBoost can effectively handle a wide range of tasks, including regression, classification, and ranking problems. It incorporates regularization techniques to control overfitting and provides flexibility in model tuning, allowing users to adjust parameters such as learning rate, tree depth, and regularization parameters to improve performance. The equation for the XGBoost algorithm, known as XGBoostRegressor for regression tasks, is as follows:

Fig. 5 Random Forest Regression structure



$$\text{Prediction: } \hat{y} = \sum_{i=1}^n f_x(x_i) \quad (2)$$

$$\text{Loss Function: } L(\hat{y}, y) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{i=1}^n \Omega(f_x) \quad (3)$$

$$\text{Regularization Term: } \Omega(f_x) = \gamma T + \frac{1}{2} \lambda \sum_{i=1}^n w_i^2 \quad (4)$$

where $\hat{y}$ is the predicted value; $\sum_i$ denotes the summation over all the data points; f_x represents the ensemble of decision trees, x_i is the input feature vector for the i -th data point, y_i is the corresponding true value; $l(\hat{y}_i, y_i)$ is the loss function that measures the discrepancy between the predicted value and the true value; γ and λ are regularization parameters; T represents the number of leaves in the decision trees; w_i is the weight associated with the i -th leaf node (Fig. 6).

The XGBoost algorithm aims to find the optimal set of decision trees f_x that minimizes the loss function $L(\hat{y}, y)$ while incorporating regularization to control model complexity and prevent overfitting. It uses a combination of gradient descent and second-order approximation techniques to efficiently optimize the objective function and build the ensemble of decision trees. XGBoost has been widely applied in various domains, including healthcare (Mishra 2018), material science (Zou et al. 2022) due to its exceptional predictive power and scalability. It has won

numerous data science competitions and is regarded as a state-of-the-art algorithm for ensemble learning.

Artificial Neural Network (ANN)

Artificial Neural Networks (ANN), inspired by biological neural networks, are versatile machine learning models (Heaton 2015). At the core of an ANN lies the fundamental unit, an artificial neuron or node, responsible for receiving input signals, performing computations, and generating output signals. These artificial neurons are structured into layers: an input layer for data input, hidden layers for intermediate computations, and an output layer for final predictions or classifications. The connections among neurons in distinct layers are characterized by weights, which govern the degree of impact one neuron exerts on another. During the training phase, the network learns by adjusting these weights based on a specified optimization algorithm and a defined loss function, such as mean squared error or cross-entropy. This process is typically done using backpropagation, calculating the loss function's gradient concerning the weights and subsequently adjusting them as needed. They find application of ANN across diverse fields like finance (Li and Ma 2010), healthcare (Shahid et al. 2019), and earth science (Belayneh et al. 2014; Cakir et al. 2014) (Fig. 7).

Ridge Regression

Ridge Regression is a regularization technique used in linear regression to address multicollinearity and overfitting

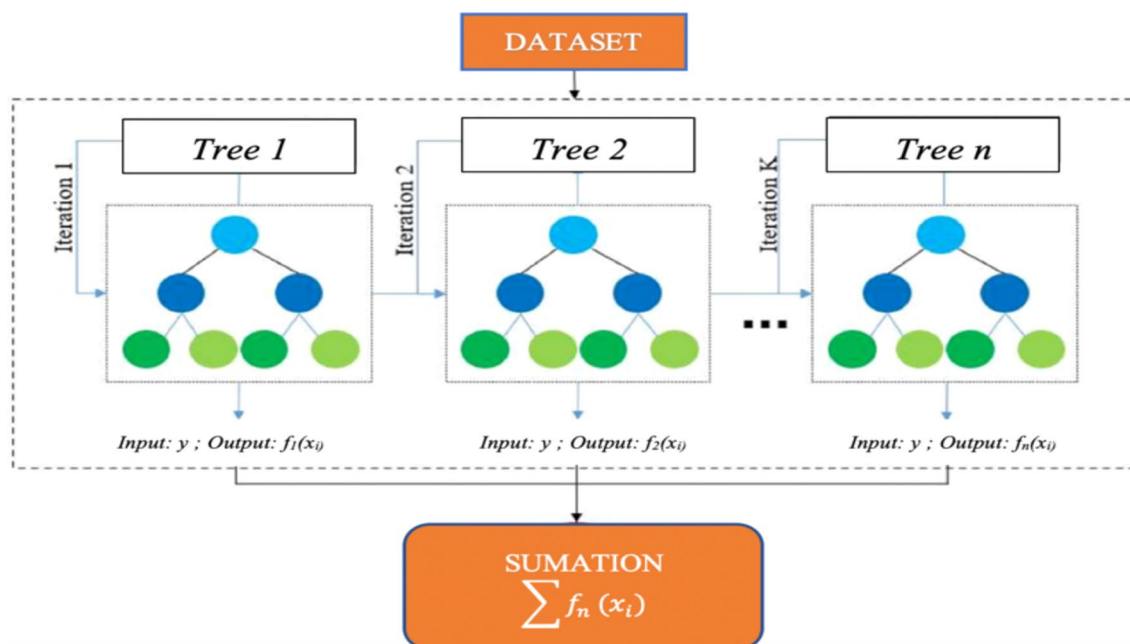
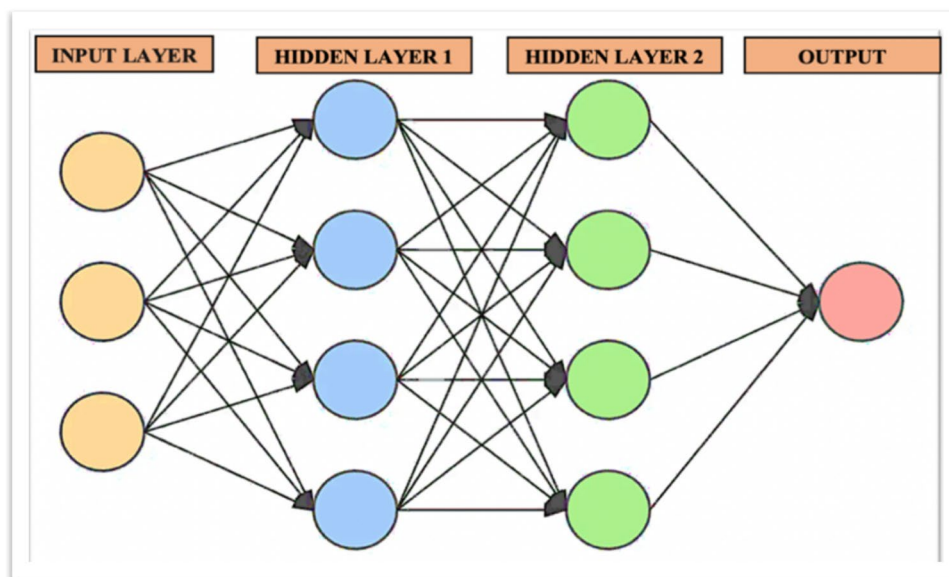


Fig. 6 Extreme Gradient Boosting (XGBoost)

Fig. 7 Artificial Neural Network (ANN) structure



issues. It adds a penalty term, known as the ridge penalty or L2 regularization term, to the error function. This penalty encourages smaller coefficient values, effectively reducing model complexity and preventing overfitting. The strength of the penalty is controlled by a hyperparameter called the regularization parameter (λ or λ). Higher values of λ result in stronger regularization and smaller coefficient magnitudes. Ridge Regression strikes a balance between capturing patterns in the data and avoiding reliance on individual data points. It shrinks the coefficients toward zero but does not eliminate them entirely, reducing the impact of multicollinearity and improving model stability.

For this research, we employ scikit-learn's ridge class to construct the ridge Regression model. We establish a grid of hyperparameter values using the `param_grid` dictionary, specifying different values for the regularization parameter α (λ -parameter) value. The GridSearch is then used to find the best parameters which will work for the Ridge Regressor.

Construction of machine learning models

The construction of machine learning models is (MLMs) a critical step in the research process. In this study, different ML models have been developed to predict salinity at Tran De station, which is near the mouth of Hau river and

has been heavily affected by saline intrusion in recent years (GSO 2016, 2020). These models were constructed using a combination of input variables and algorithmic techniques. In this study, the process of devising appropriate input combinations involves two main steps. Firstly, several input combinations were tested using different variables based on their correlation coefficients with salinity at Tran De station. Six input combinations were created, starting with the variable with the highest correlation coefficient and gradually adding variables in a stepwise manner until all variables were included. Statistical indexes were used to assess the performance of each combination in the test set, ultimately identifying the most effective combination.

Table 4 displays the coefficient of determination (R^2), which quantifies the strength and direction of the relationship between input variables and salinity at Tran De station. A higher R^2 value indicates a stronger relationship and implies that a larger proportion of the variance in salinity can be explained by the corresponding input variable.

Table 5 presents various input variable combinations used in the modeling process. These scenarios represent different configurations of input variables that were tested to assess their impact on the predictive performance of the models.

In addition to selecting the combination parameters scenario, we also aim to determine the lead time for conducting the salinity prediction model. Determining the lead time for

Table 4 Coefficient of determination (R^2) between the input variables and salinity concentration at Tran De station

Input variables	Salinity concentration at Long Phu station	Salinity concentration at Dai Ngai station	Salinity concentration at An Lac Tay station	Salinity concentration at Soc Trang station	Water level	Discharge
Correlation	0.93	0.78	0.49	0.65	0.25	−0.39

Table 5 Scenarios input combination used in the modeling

No	SCENARIOS input combination
1	Salinity at Tran De and Long Phu station
2	Salinity at Tran De, Long Phu and Dai Ngai stations
3	Salinity at Tran De, Long Phu, Dai Ngai, Soc Trang stations
4	Salinity at Tran De, Long Phu, Dai Ngai, Soc Trang, An Lac Tay stations
5	Salinity at Tran De, Long Phu, Dai Ngai, Soc Trang, An Lac Tay stations and Discharge
6	Salinity at Tran De, Long Phu, Dai Ngai, Soc Trang, An Lac Tay stations, Discharge, Water level

simulation in the salinity prediction model involves sequentially changing the previous forecast time and evaluating the model's efficiency (Section “[Model evaluation](#)”). This process is performed by assessing the correlation between salinity concentrations at the current Tran De station and its own past time periods. The Partial Autocorrelation Function (PACF) is a tool used in time-series analysis to measure the correlation between data points in a time series while accounting for the influence of intermediate values. It allows us to determine the degree of interaction between data points in the time series and helps in building an efficient predictive model. Utilizing the PACF, we can identify the optimal time intervals for the salinity prediction model (Wang 2013).

Model evaluation

The evaluation of the model's performance was conducted using five statistical criteria including the correlation of determination (R^2) (Willmott 1982); the root-mean-square error (RMSE) (Maheswaran and Khosa 2012); the mean absolute error (MAE), and the mean square error (MSE) (Willmott et al. 2005).

$$R^2 = 1 - \frac{\sum_{i=1}^n (O_i - P_i)^2}{\sum_i (O_i - \bar{O})^2} \quad (5)$$

$$\text{RMSE} = \left[\sqrt{\frac{\sum_{i=1}^n (O_i - P_i)^2}{n}} \right] \quad (6)$$

$$\text{MAE} = \frac{\sum_{i=1}^n (O_i - P_i)}{n} \quad (7)$$

$$\text{MSE} = \frac{\sum_{i=1}^n (O_i - P_i)^2}{n} \quad (8)$$

where P_i is the forecast value; O_i is the observation value; $\bar{O}$ is the average observation values; $\bar{P}$ is the average forecast values; n is the total number of observation.

Furthermore, the Taylor diagram was employed to visually compare the model's performance. The Taylor diagram provides a two-dimensional plot where a single point represents the correlation coefficient, the root-mean-square difference (RMSD), and the ratio of standard deviations between the two variables (SD). This diagram is especially useful for comparing different models and monitoring their overall performance over time (Taylor 2001).

Results

Identifying the optimal timing for predicting salinity

Autocorrelation and partial autocorrelation plots were employed to determine the optimal timing for predicting salinity. These plots provide valuable insights into the correlation patterns between different time periods and assist in identifying the most suitable time intervals for salinity prediction. The results of autocorrelation and partial autocorrelation for the salinity variable are depicted in the figure below.

The autocorrelation plot reveals that there is minimal difference in the correlation value between the salinity values at the previous time period and the current time period ($t=0$). It is evident that the correlation of salinity values at different time points shows cyclical variation, with each cycle repeating after 7 time steps. Moreover, the partial autocorrelation plot, which excludes the influence of other factors, demonstrates a distinct correlation between the salinity values at the previous forecast time and the current time, suggesting that this cycle is likely to last up to 8 time steps. Based on these findings, we will utilize different time steps, with each time step (i) representing a 2-h interval. The time steps $i=1$ to 8 correspond to predicted time intervals ranging from 2 to 16 h before the current time, used to predict the salinity at the Tran De station. These specific time intervals have been selected based on the notable correlations observed in the partial autocorrelation chart.

Determining the lead time for prediction involves comparing and evaluating the performance of each model at different forecasting periods (2–16 h) during the training

stage. The performance evaluation results for each model are presented in Table 6.

Based on the information provided in Table 6 and Fig. 8, although there is no statistically significant difference observed among the forecasting periods, it is evident that the models perform more efficiently when forecasting up to 16 h ($i=8$) in advance compared to other time periods. The models demonstrate higher efficiency, with R^2 values ranging from 0.87 to 0.97, RMSE values between 0.79 and 1.74 for all models. The results suggest that the majority of models are highly effective in predicting salinity within a 16-h advance forecast period. Additionally, early notification is crucial in providing information to the community, enabling them to take timely measures to cope with saltwater intrusion. Consequently, a lead time of 16 h ($i=8$) appears to

strike an optimal balance between accuracy and prediction interval for the research models, making it a valuable tool for informing and assisting the community in dealing with saltwater intrusion challenges.

The analysis results outlined above have led to the identification of a specific combination of input variables, which is detailed in Table 7.

Performance of the Random Forest Regression algorithm

Table 8 presents the statistical measurements of the models evaluated using both the training and testing datasets. It offers an overview of the Random Forest algorithm's performance during both the training and testing phases.

Table 6 Model performance of the six machine learning algorithms in identifying the optimal timing for predicting salinity

Scenarios		Lead time (h)							
		2 ($i=1$)	4 ($i=2$)	6 ($i=3$)	8 ($i=4$)	10 ($i=5$)	12 ($i=6$)	14 ($i=7$)	16 ($i=8$)
Random Forest Regression	R^2	0.92	0.92	0.89	0.92	0.97	0.97	0.92	0.97
	RMSE	0.04	0.03	0.04	0.04	0.82	0.82	0.03	0.79
Support Vector Regression	R^2	0.94	0.95	0.89	0.95	0.96	0.97	0.95	0.97
	RMSE	0.03	0.03	0.04	0.03	0.94	0.89	0.03	0.85
Long Short-Term Memory	R^2	0.87	0.83	0.80	0.84	0.88	0.87	0.81	0.87
	RMSE	0.04	0.05	0.05	0.05	1.71	1.8	0.05	1.74
Artificial Neural Network	R^2	0.93	0.88	0.86	0.93	0.96	0.95	0.94	0.96
	RMSE	0.03	0.04	0.04	1.03	1.02	1.05	0.03	1.02
Extreme Gradient Boosting	R^2	0.93	0.94	0.88	0.93	0.97	0.97	0.93	0.97
	RMSE	0.03	0.03	0.04	0.03	0.85	0.84	0.03	0.82
Ridge Regression	R^2	0.93	0.95	0.86	0.94	0.95	0.95	0.95	0.95
	RMSE	0.03	0.03	0.04	0.03	1.08	1.07	0.02	1.05

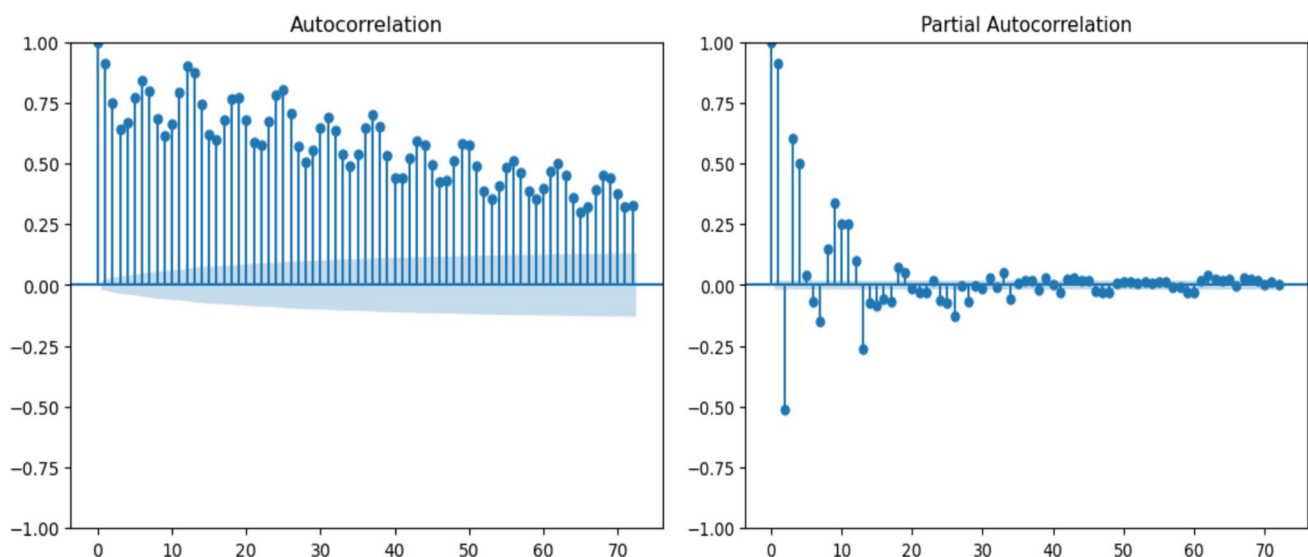


Fig. 8 Autocorrelation and partial autocorrelation plots for determining the lead time for predicting salinity at Tran De station

The results show that for most scenarios throughout both the training and testing stages, R^2 value is typically greater than 0.96. RMSE varies between 0.79 and 0.94 during training phase and fluctuates around 1.0 during testing phase. MSE for the training phase ranges from 0.62 to 0.89, while during the testing phase, it ranges from 0.97 to 1.11. Additionally, MAE for the training and testing stages ranges from 0.48 to 0.73. Among the scenarios considered, scenario 6 demonstrates the highest performance in predicting salinity at Tran De station. With R^2 value of 0.97, RMSE of 0.79, MAE of 0.48, and MSE of 0.62, scenario 6 outperforms the other scenarios in accurately predicting salinity levels. These results suggest that scenario 6 provides a better fit to the observed data and exhibits a smaller deviation from the true values compared to the other scenarios.

Performance of the SVR algorithm

The model statistics for training and testing phases are presented in Table 9. The results indicate that R^2 value, which measures the goodness of fit, consistently exceeds 0.9 for most cases during both the training and testing phases; this suggests a strong linear relationship between the predicted and actual values of salinity. In the testing phase, RMSE ranges from 0.89 to 0.91, MSE in the training phase ranges from 0.79 to 0.83, and in the testing phase, it ranges from 0.73 to 1.37. MAE, which quantifies the average absolute difference between the predicted and observed values and fluctuates between 0.51 and 0.83 for both the training and testing periods. The R^2 , RMSE, MSE, and MAE values indicate that the model performs well overall and is able to forecast salinity levels with accuracy.

Among the scenarios considered, scenario 6 shows the highest performance in predicting salinity at Tran De station.

Table 7 Scenarios input combination in predicting salinity

No	Scenarios input combination (t-i)
1	Salinity at Tran De station (t-8) and Long Phu station (t-8)
2	Salinity at Tran De station (t-8), Long Phu station (t-8) and Dai Ngai stations (t-8)
3	Salinity at Tran De station (t-8), Long Phu station (t-8), Dai Ngai station (t-8) and Soc Trang stations (t-8)
4	Salinity at Tran De station (t-8), Long Phu station (t-8), Dai Ngai station (t-8), Soc Trang station (t-8), An Lac Tay stations (t-8)
5	Salinity at Tran De station (t-8), Long Phu station (t-8), Dai Ngai station (t-8), Soc Trang station (t-8), An Lac Tay stations (t-8), Discharge (t-8)
6	Salinity at Tran De station (t-8), Long Phu station (t-8), Dai Ngai station (t-8), Soc Trang station (t-8), An Lac Tay stations (t-8), Discharge(t-8), Water level(t-8)

Table 8 Model performance of the RFR algorithm during training and testing phase

Scenarios	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
S1	0.62	0.89	0.96	0.94	0.72	1.08	0.96	1.04
S2	0.61	0.87	0.96	0.93	0.73	1.09	0.96	1.05
S3	0.59	0.83	0.97	0.91	0.73	1.08	0.96	1.04
S4	0.5	0.67	0.97	0.82	0.73	1.09	0.96	1.04
S5	0.5	0.67	0.97	0.82	0.68	0.97	0.98	0.98
S6	0.48	0.62	0.97	0.79	0.65	0.91	0.97	0.95

Table 9 Model performance of the SVR algorithm during training and testing phase

Scenarios	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
S1	0.79	1.37	0.9	1.17226	0.61	0.80	0.97	0.89
S2	0.76	1.30	0.95	1.14	0.61	0.79	0.97	0.89
S3	0.74	1.26	0.95	1.12	0.60	0.79	0.97	0.89
S4	0.73	1.25	0.95	1.12	0.62	0.81	0.97	0.90
S5	0.57	0.88	0.96	0.94	0.62	0.83	0.97	0.91
S6	0.51	0.73	0.97	0.85	0.6	0.79	0.97	0.89

This is supported by its impressive R^2 value of 0.97, indicating a strong correlation between the predicted and actual salinity values. RMSE of 0.89 signifies a small average deviation of the predictions from the observed values. Additionally, MAE of 0.6 and MSE of 0.79 further demonstrate the accuracy of scenario 6 in predicting salinity. When compared to the other scenarios, scenario 6 consistently exhibits a closer fit to the observed data and demonstrates a smaller deviation from the true values. This suggests that Scenario 6 provides more reliable and accurate predictions of salinity levels.

Performance of the LSTM algorithms

The performance of the LSTM algorithm is presented in Table 10. The LSTM algorithm exhibits a good linear relationship between the predicted and observed values, as indicated by the R^2 values ranging from 0.82 to 0.87 during the training phase and 0.88 to 0.9 during the testing phases. Additionally, the algorithm achieves a RMSE fluctuation of 2.0, indicating a low average magnitude of errors between the predicted and actual data. The model's improved predictive potential is further demonstrated by MAE and MSE indexes, which are approximately 1.5 and range from 3.04 to 4.3, respectively.

The findings also show that scenario 6 exhibits the highest salinity prediction capability among the scenarios examined. It achieves a R^2 value of 0.9, indicating a strong correlation between the predicted and actual salinity measurements. Furthermore, scenario 6 exhibits low values for root-mean-square error (RMSE), mean absolute error (MAE), and mean

squared error (MSE) during the testing phase, with values of 1.75, 3.07, and 1.36 respectively. These metrics signify a small average deviation of the predictions from the observed values, highlighting the accuracy of Scenario 6 in predicting salinity. Hence, Scenario 6 consistently provides a closer match to the observed data and exhibits a smaller deviation from the true values compared to the other scenarios, making its salinity-level estimates more reliable and precise.

Performance of the Artificial Neural Network (ANN) algorithm

Table 11 presents a summary of the effectiveness of the ANN technique in predicting salinity. It showcases the statistical measurements of the models evaluated using both the train and test datasets. The statistical indicators indicate that the ANN model is well-suited for salinity prediction, with R^2 value consistently around 0.95. RMSE values range from 0.94 to 1.08, MAE varies from 0.7 to 0.82, and MSE ranges from 0.78 to 1.39. Although there is no significant difference among the statistical indicators across the scenarios, a closer examination of the detailed data suggests that scenario 1 performs slightly better than the others.

Performance of the Extreme Gradient Boosting (XGBoost) algorithm

Table 12 provides an overview of the XGBoost algorithm's performance in forecasting salinity. The table provides a statistical analysis of the models that were examined using the train and test sets of data. According to the statistical

Table 10 Model performance of the LSTM algorithm during training and testing phase

Scenarios	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
S1	1.64	4.3	0.82	2.07	1.39	3.16	0.89	1.78
S2	1.58	4.06	0.83	2.02	1.34	2.99	0.91	1.73
S3	1.47	3.61	0.85	1.9002	1.31	2.84	0.90	1.69
S4	1.52	3.86	0.84	1.96	1.51	3.76	0.88	1.94
S5	1.52	3.86	0.84	1.96	1.51	3.76	0.88	1.94
S6	1.31	3.04	0.87	1.74	1.36	3.07	0.9	1.75

Table 11 Model performance of the ANN algorithms during training and testing phase

Scenarios	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
S1	0.75	1.25	0.95	1.12013	0.58	0.78	0.97	0.88
S2	0.73	1.17	0.95	1.08	0.59	0.79	0.97	0.89
S3	0.82	1.39	0.94	1.18	0.67	0.91	0.97	0.95
S4	0.8	1.3	0.95	1.14	0.69	0.98	0.97	0.99
S5	0.72	1.13	0.95	1.07	0.77	1.12	0.96	1.05
S6	0.70	1.03	0.96	1.02	0.79	1.17	0.96	1.08

Table 12 Model performance of the XGBoost algorithm during training and testing phase

Scenarios	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
S1	0.69	1.01	0.96	1.01	0.73	1.08	0.96	1.04
S2	0.67	0.96	0.96	0.98	0.73	1.08	0.96	1.04
S3	0.66	0.94	0.96	0.97	0.75	1.09	0.96	1.04
S4	0.67	0.94	0.96	0.97	0.76	1.13	0.96	1.06
S5	0.57	0.74	0.97	0.86	0.71	1.01	0.97	1.00
S6	0.54	0.68	0.97	0.82	0.7	1.17	0.97	0.99

indicators, the XGBoost model demonstrates effectiveness in forecasting salinity. The R^2 values, which vary around 0.96, indicate a significant association between the anticipated and actual salinity values. The RMSE values varying around 1.0 represent the average amount of errors between the anticipated and actual values. MAE values fluctuating around 0.7 show the average absolute difference between the projected and actual values. Additionally, the MSE values range from 0.68 to 1.17, indicating the average squared difference between the predicted and actual values.

Based on the statistical indicators, scenario 6 is identified as the scenario with a more effective combination of input variables compared to the other scenarios. The performance of scenario 6 is supported by the higher values of the statistical indicators, indicating a stronger association between the predicted and actual values, lower errors in prediction (RMSE = 1.09, MAE = 0.79), and better accuracy in forecasting ($R^2 = 0.96$). Therefore, scenario 6 demonstrates a more effective combination of input variables for predicting salinity compared to the other scenarios.

Performance of the Ridge Regression algorithm

In Table 13, a summary of the model statistics of the Ridge Regression algorithm is presented. The outcomes indicate that the Ridge Regression algorithm performs well in predicting salinity, with high R^2 values and relatively low MSE and RMSE values in both the training and testing phases. Scenario 6 is the scenario with a more efficient combination of input variables compared to the other scenarios, according to the statistical indicators. Higher values of the statistical

indicators, which show a stronger correlation between projected and actual values, less prediction errors (RMSE = 0.9, MAE = 0.82), and better forecasting accuracy ($R^2 = 0.97$), confirm scenario 6's performance. In comparison with the other situations, scenario 6 shows a stronger combination of input variables for predicting salinity.

Discussion

Assess the performance analysis of various machine learning algorithms for salinity prediction

Besides evaluating the effectiveness of each model in simulating salinity, comparing the efficiency between models is also crucial in determining the algorithm's suitability in different contexts. In this study, scenario 6, which involves a combination of inputs such as salinity at Tran De, Long Phu, Dai Ngai, Soc Trang stations, discharge, and water level, will be selected to explore the differences between models. The performance evaluation is based on several statistical metrics, including R^2 , RMSE, MAE, and MSE. These metrics provide insights into the accuracy, precision, and overall performance of the models in predicting salinity. The performance evaluation results of the models are provided in Table 14, Figs. 9a, b, 10, and 11.

The results provided in Table 14 and Fig. S1 (Appendix A) reveal that there is no statistically significant difference between the models. However, upon closer examination, it becomes evident that the RFR model outperforms the other models in terms of predictive simulation efficiency

Table 13 Performance of the Ridge Regression algorithm during training, testing phases

Scenarios	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
S1	0.9	1.6	0.93	1.26	0.64	0.86	0.97	0.93
S2	0.89	1.57	0.94	1.25	0.64	0.85	0.97	0.92
S3	0.89	1.56	0.94	1.25	0.65	0.87	0.97	0.93
S4	0.89	1.56	0.94	1.25	0.65	0.87	0.97	0.93
S5	0.77	1.27	0.95	1.13	0.65	0.87	0.97	0.93
S6	0.72	1.1	0.95	1.05	0.64	0.82	0.97	0.9

Table 14 Performance evaluation of different machine learning algorithms

	Training				Testing			
	MAE	MSE	R^2	RMSE	MAE	MSE	R^2	RMSE
Random Forest Regression	0.48	0.62	0.97	0.79	0.65	0.91	0.97	0.95
Support Vector Regression	0.51	0.73	0.97	0.85	0.6	0.79	0.97	0.89
Artificial Neural Network	0.70	1.03	0.96	1.02	0.79	1.17	0.96	1.08
Long Short-Term Memory	1.31	3.04	0.87	1.74	1.36	3.07	0.9	1.75
Extreme Gradient Boosting	0.54	0.68	0.97	0.82	0.7	1.17	0.97	0.99
Ridge Regression	0.72	1.1	0.95	1.05	0.64	0.82	0.97	0.9

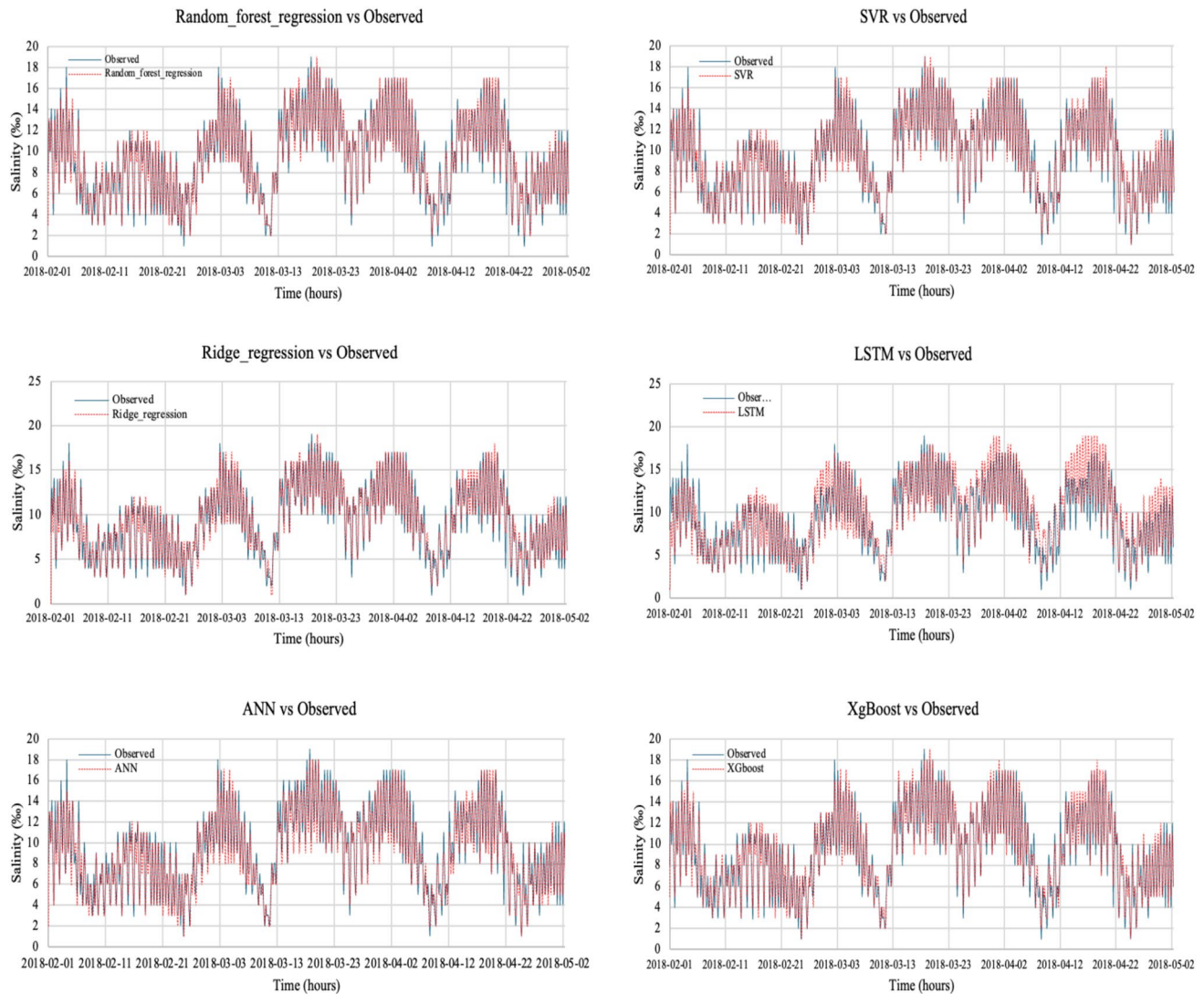


Fig. 9 **a** Predicted and observed salinity for the SVR, RFR, LSTM, ANN, XGBoost, and Ridge Regression models in 2018, during the training phase; **b** Predicted and observed salinity for the SVR, RFR,

LSTM, ANN, XGBoost, and Ridge Regression models in 2020, during the testing phase

and stability, as reflected in its R^2 , MSE, MAE, and RMSE metrics. The RFR model exhibits an R^2 value exceeding 0.97 for both the training and testing phases. Additionally, furthermore, the training phase exhibits an MSE of 0.62

and an RMSE of 0.79, while the testing phase shows an MSE of 0.91 and an RMSE of 0.95. These metrics indicate the SVR model's superior performance and accuracy in predicting salinity intrusion compared to other models,

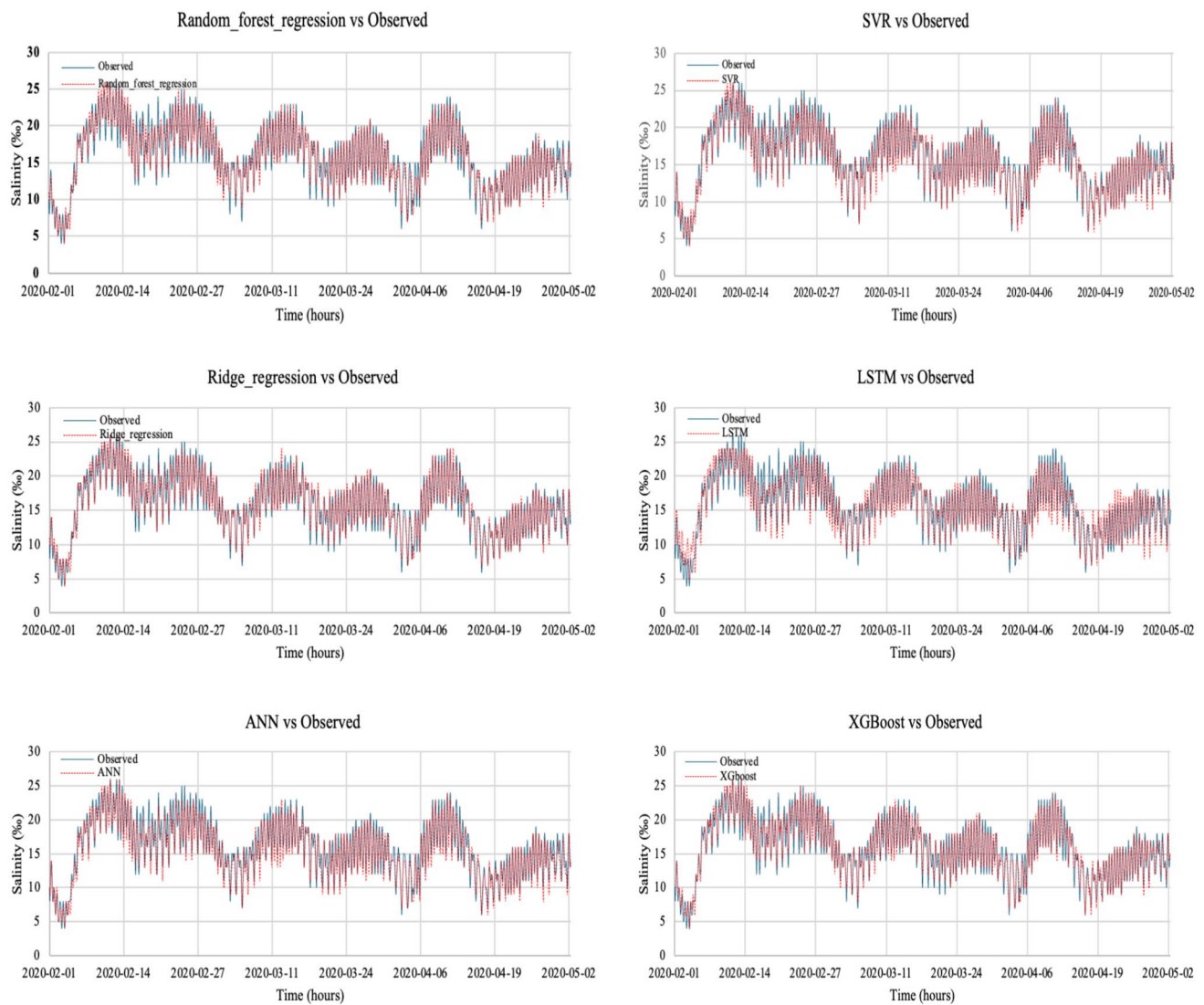


Fig. 9 (continued)

whereas LSTM performed lower than the other models (ANN, RFR, XGBoost, Ridge Regression, and SVR) in predicting salinity intrusion.

To provide a more specific and detailed comparison of the machine learning models' performance in predicting salinity levels, data from 2018 (training phase) and 2020 (testing phase) were randomly selected for analysis and are presented in Fig. 9a and b, respectively. Based on these figures, no significant discrepancies are observed between predicted and observed salinity levels in most models during both the training and testing phases. This reflects the models' effectiveness in predicting salinity levels. Additionally, when comparing the models' performance, the LSTM model consistently exhibits larger gaps between predicted and observed salinity levels compared to the other models.

Figure 10 illustrates the distribution of predicted salinity concentrations among the models. Notably, the RF and SVR models exhibit undeniable accuracy, closely aligning with the observed data. However, the LSTM model demonstrates a significant deviation from the observed data. Additionally, when considering median and quartile values, although there are slight variations between these models and the observed data, both the RFR and SVR models show a high level of simulation efficiency. Following these models, the ANN, XGBoost, and Ridge Regression models also show relatively high simulation efficiency.

The Taylor diagram (Fig. 11) displays the correlation coefficient (R), standard deviation (SD), and centered root-mean-square difference (RMSD) indexes of the six models during the testing phase. The models exhibit correlation coefficient (R) values within the range of 0.95 to 0.99,

Fig. 10 Distribution of salinity concentrations of machine learning models in the testing phase

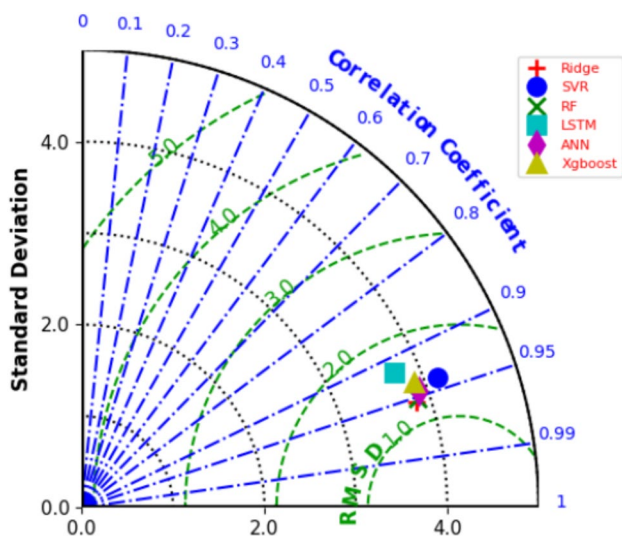
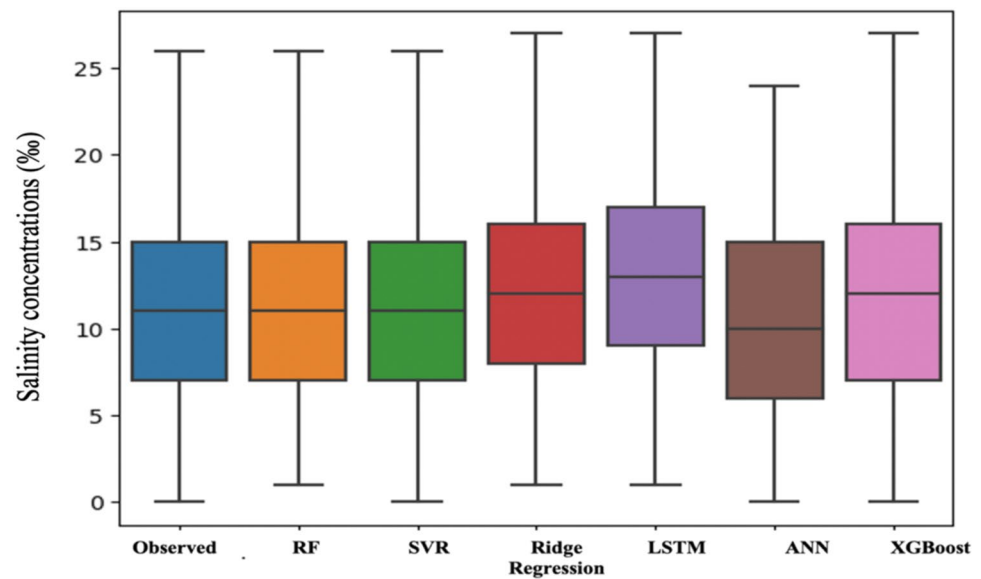


Fig. 11 Taylor diagram illustrates statistics (correlation coefficient, standard deviation, and centered root-mean-square difference) between salinity at Tran De and salinity of machine learning models in testing phase

indicating good overall simulation performance. However, the LSTM model falls below the threshold of 0.95. When examining the Taylor diagram, it provides additional information that aligns with the findings in Table 14. Although the simulation performance values do not show a significant difference, the RFR and SVR models still emerge as the best-fit models for simulating salinity in the study area.

Taking into account the LSTM model, it performed worse than the other models (ANN, RFR, XGBoost, Ridge Regression, and SVR) in predicting salinity intrusion. This lower performance can be attributed to various factors, including model complexity, insufficient training data, and dataset

characteristics (Houdt et al. 2020). As shown in Table 10, the R^2 value for the training phase of the LSTM model is 0.88, whereas it increases to 0.90 for the testing phase. This suggests that the model's performance improves during testing, indicating a potential issue with the amount or quality of training data. Therefore, the lower performance of the LSTM model can be attributed to insufficient training data, which limits its ability to effectively learn and generalize patterns.

The SVR model is highly effective in capturing and demonstrating the relationship between input variables during both the training and verification stages. The performance metrics of the SVR model in this study are impressive, with a typical R^2 value exceeding 0.97 during the training and testing phases. Additionally, RMSE values range from 0.85 to 0.89, and the MSE ranges from 0.72 to 0.79 during the training and testing phases. These metrics indicate the model's ability to accurately predict salinity intrusion and its stability across diverse datasets and scenarios.

The ANN model exhibits high simulation efficiency during the training stage but demonstrates relatively poorer performance during the testing stage. This can be attributed to the inherent noise in the ANN model, which arises from its dependence on learning and memory. Each learning iteration in the ANN model produces different errors, leading to higher levels of noise when the data are not uniform or when there are insufficient training data. However, despite its noise, the ANN model proves to be effective in predicting salinity when used in combination with different input variables. The ANN model demonstrates its effectiveness in capturing the underlying patterns and relationships within the salinity data, as evidenced by the values of the statistical indexes. The high R^2 value of over 0.96 indicates that the ANN model explains a significant portion of the variance in the salinity data, suggesting a strong relationship

between the input variables and salinity values. The RMSE of 1.01 for the training phase and 1.08 for the testing phase indicates the accuracy of the ANN model's predictions. A lower RMSE value indicates that the predicted salinity values are closer to the observed values, signifying a higher level of precision in the model's performance. Additionally, the MAE and MSE, ranging from 0.7 to 1.17, further demonstrate the accuracy and precision of the ANN model. These metrics quantify the average magnitude of the errors in the model's predictions and indicate that the ANN model performs well in minimizing errors and providing reliable predictions of salinity.

The XGBoost model is a powerful and widely used tool for regression tasks, including predictive modeling and time-series forecasting (Abu-Rmileh 2020). The evaluation results confirm the strength of the XGBoost model, with R^2 values above 0.97 for both the training and testing periods. This indicates a high level of explained variance by the model in relation to the observed salinity data. Additionally, the evaluation metrics such as RMSE, MAE, and MSE provide further evidence of the model's effectiveness. The range of these metrics falls between 0.54 and 0.99, indicating relatively low levels of prediction error.

The Ridge Regression model, known for its ability to model linear relationships between variables, yielded statistical parameters in this study, including an R^2 of 0.97, RMSE of 0.91, MAE of 0.63, and MSE of 0.82 during the testing phase. These results demonstrate its suitability for simulating salinity in the study area, indicating its effectiveness in capturing linear relationships between variables and providing accurate predictions.

The issue of saltwater intrusion has gained significant attention due to its impact on coastal communities and livelihoods. Researchers have developed various models to predict salinity in different regions. Nicolas et al. (2023) aimed to predict salinity using machine learning algorithms such as MLP, SVR, and RFR. Their research findings demonstrated that despite the simplicity of the input data, these algorithms effectively captured the seasonal and semi-diurnal variations of sea surface salinity, including tide-induced modulation and low salinity events during winter. Among the algorithms tested, Support Vector Regression (SVR) produced the most accurate estimates. Furthermore, Tran et al. (2022) conducted research on salinity prediction in the Ham Luong River using machine learning models such as KNNR, SVR, LSTM, and RFR. The input variable considered in the study was the salinity of monitoring stations. The findings suggested that, within the study area, the LSTM model outperformed other machine learning models.

Our research findings are consistent with the general trends observed in previous studies on salinity prediction (Alizadeh et al. 2018; Rath et al. 2017; Zhou et al. 2021; Hu et al. 2019). Specifically, our study demonstrates that

both ANN and SVR exhibit exceptional capabilities in handling complex, nonlinear, and multidimensional environmental datasets. These results align with Alizadeh et al. (2018), which highlighted the superiority of these models in addressing intricate relationships within environmental data. Furthermore, our findings regarding the minimal influence of flow at the Can Tho station on predicted salinity concentrations are consistent with Zhou et al. (2021) research on the application of machine learning algorithms to similar challenges.

These findings collectively highlight the potential of machine learning models in predicting and managing salinity levels, providing valuable insights and early warning solutions for addressing saltwater intrusion issues. This can significantly contribute to mitigating the challenges associated with salinity intrusion in the VMD and enhance the overall understanding of salinity prediction in coastal areas.

Extending model validation to other condition scenarios

To comprehensively evaluate the performance and reliability of the forecasting models, it is crucial to validate them under real-world conditions. In this study, we conducted salinity prediction at the Tran De station for the year 2023. The forecasting results are presented in Table 15 and Fig. 12. The results from Table 15 show that the models performed well under the new conditions, with low MSE values across all models. The R^2 values range from 0.81 to 0.94; however, the performance of the LSTM model remains lower compared to the other models. The RMSE values range from 0.02 to 0.05, while the MSE values vary from 0.02 to 0.04. This indicates that the models are well-adapted to the new conditions and maintain high effectiveness in predicting salinity in the study area.

Additionally, the data shown in Fig. 12 and Fig. S2 (Appendix A) represent the time-series comparison between predicted and actual salinity levels. Specifically, Fig. 12 illustrates salinity concentrations for April 2023 and Fig. S2 represents salinity variations throughout the entire validation period. This chart not only illustrates the variation of

Table 15 Performance of machine learning algorithms in salinity prediction

	MAE	MSE	R^2	RMSE
Random Forest Regression	0.027	0.00	0.92	0.04
Support Vector Regression	0.02	0.00	0.94	0.03
Artificial Neural Network	0.02	0.00	0.94	0.03
Long Short-Term Memory	0.04	0.00	0.84	0.05
Extreme Gradient Boosting	0.02	0.00	0.94	0.03
Ridge Regression	0.02	0.00	0.95	0.03

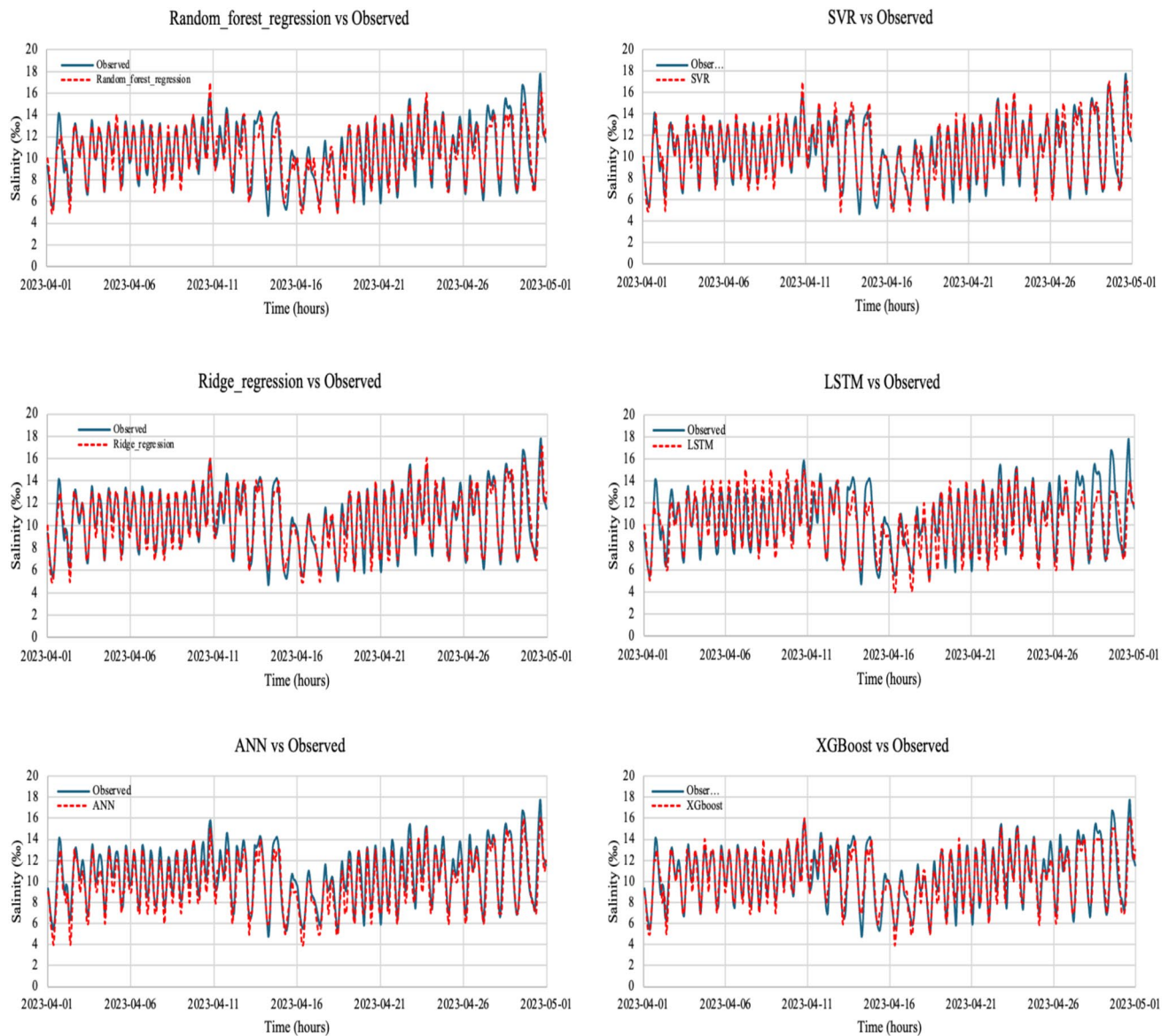


Fig. 12 Comparison of salinity prediction results with observed salinity in April 2023 during the validation stage of SVR, RFR, LSTM, ANN, XGBoost, and Ridge Regression models

salinity levels over different periods but also demonstrates the degree of alignment between the predicted and actual values, thereby reflecting the effectiveness and accuracy of the forecasting models under real-world conditions.

Besides, the salinity data detailed for April 2023 reveal that, consistent with the calibration and testing phases, the LSTM model exhibits larger deviations between predicted and observed salinity levels compared to the other models.

Limitation of research

The main objective of this study was to develop and explore the potential of machine learning algorithms for

salinity prediction in the Mekong Delta region of Vietnam. Throughout the research process, we achieved our goal by applying appropriate research methods and analyzing relevant data. Nonetheless, it is essential to recognize the constraints inherent in our study.

Firstly, salinity concentration is influenced by various factors (Liu et al. 2014; Shaha et al. 2013; Byung and Seung-Buhm 2013; Ralston et al. 2008a, 2008b; Uncles and Stephens 2011; Melesse et al. 2020; Pereira et al. 2019; Singh et al. 2020). However, due to data limitations, we could only consider the impact of salinity, discharge, and water level in our salinity forecasts. It is important to recognize that other factors may also play a significant

role in salinity prediction, and future studies should aim to incorporate a broader range of variables such as precipitation, tidal levels, and wind, among others, to enhance the accuracy of the predictions.

Secondly, in determining the optimal forecasting time, our approach focused primarily on the relationship between salinity and salinity variables from the previous hour. We did not extensively explore the optimization of forecasting time for each input parameter. Future studies should consider a more comprehensive approach to determine the optimal forecasting time for individual variables, taking into account their characteristics and dynamics.

Additionally, it is essential to highlight that our research results are limited to the regional context of the Mekong Delta. To enhance the significance and applicability of our findings, it would be beneficial to extend the study to different regions and consider a global perspective. This broader perspective would provide valuable insights into the dynamics of saline intrusion and facilitate discussions on potential cooperation, knowledge sharing, and international policy interventions to address salinity intrusion and its consequences.

We acknowledge these limitations and intend to address them in future research. By incorporating a wider range of variables, optimizing forecasting time, and considering diverse regions, we can enhance the reliability, validity, and practical implications of our research findings.

Conclusion

In this study, we employed a thorough and interdisciplinary methodology to forecast salinity levels in the Soc Trang region. We combined knowledge from environmental science, hydrological processes, earth systems, and computer science to enhance the accuracy of salinity prediction. The principal objective of the investigation was to develop machine learning algorithms, including RFR, SVR, RR, ANN, LSTM, and XGBoost, for short-term salinity prediction and to compare the effectiveness of these different algorithms in predicting salinity. The performance of the models was assessed using statistical indices and graphical comparisons.

The research findings demonstrate the effectiveness of machine learning algorithms in short-term salinity prediction. Although no statistically significant difference was observed, the evaluation of statistical coefficients indicates that the RFR model exhibits the best performance in salinity prediction, followed by the SVR, ANN, XGBoost, Ridge Regression, and Long Short-Term Memory (LSTM) models.

Furthermore, when considering different scenarios of input parameter combinations, the RFR, XGBoost, and Ridge Regression models prove highly effective in predicting

salinity in the study area, especially in scenario 6. This scenario includes the combination of salinity variables from Tram Tran De, Long Phu, Dai Ngai, Soc Trang, and An Lac Tay stations, as well as water level and discharge. Similarly, the SVR model achieves high accuracy in predicting salinity in scenario 2, where the input variables include salinity from Tran De, Long Phu, and Dai Ngai stations.

On the other hand, the LSTM and ANN models demonstrate effectiveness in predicting saline intrusion in scenario 3, incorporating input variables of salinity from Tran De, Long Phu, Dai Ngai, and Soc Trang stations. The research also identifies the forecasting time of up to 16 h ($i=8$) as suitable for salinity prediction in the study area, providing an initial foundation for further investigations into saltwater intrusion.

These findings offer valuable insights into the effectiveness of various machine learning algorithms for salinity prediction and can guide the selection of suitable models in future studies or practical applications. Subsequent research can further build upon this foundation by developing machine learning models to investigate the spread of salinity in the study area, addressing saltwater intrusion prediction, and tackling similar mathematical challenges using advanced machine learning algorithms. Such endeavors have the potential to be transformative developments in the field of applying machine learning to hydrological studies.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical standards This article does not contain any studies with human participants or animals performed by any of the authors.

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References

- Abu-Rmieleh A (2020) The multiple faces of 'feature importance' in XGBoost. *Towards Data Sci*
- Aditya C, Akash S, Ayush K, Karan D, Neeraj K (2020) Short term wind power forecasting using machine learning techniques. *J Stat Manag Syst* 23(1):145–156. <https://doi.org/10.1080/09720510.2020.1721632>
- Alizadeh MJ, Kavianpour M, Danesh M, Adolf J, Shamshirband S, Chau K (2018) Effect of river flow on the quality of estuarine and coastal waters using machine learning models. *Eng Appl Comput Fluid Mech* 12:810–823
- Ardabili S, Mosavi A, Varkonyi-Koczy A (2019) Systematic review of deep learning and machine learning models in biofuels research. In: *International conference on global research and education*. <https://doi.org/10.20944/preprints201908.0179.v1>
- Belayneh A, Adamowski J, Khalil B, Ozga-Zielinski B (2014) Long-term SPI drought forecasting in the Awash River Basin in Ethiopia using wavelet neural networks and wavelet support vector regression models. *J Hydrol* 508:418–429
- Bergstra J, Bengio Y (2012) Random search for hyper-parameter optimization. *J Mach Learn Res* 13(2):281–305. <https://doi.org/10.1016/j.chemolab.2011.12.002>
- Breiman L (2001) Random forests. *Mach Learn* 45:5–32. <https://doi.org/10.1023/A:1010933404324>
- Byung Y, Seung-Buhm W (2013) Correlation between freshwater discharge and salinity intrusion in the Han River Estuary, South Korea. *J Coast Res* 65(sp2):1247–1252. <https://doi.org/10.2112/S165-211.1>
- Cakir Y, Kirici M, Gunes EO (2014) Yield prediction of wheat in south-east region of Turkey by using artificial neural networks. In: *2014 the 3rd international conference on agro-geoinformatics, agro-geoinformatics*. <https://doi.org/10.1109/Agro-Geoinformatics.2014.6910609>
- Cameron WM, Pritchard DW (1963) Estuaries, in the sea: ideas and observations on progress in the study of the seas. In: Hill MN (ed) *The composition of sea-water, comparative and descriptive oceanography*, Wiley, New York, vol 2, pp 306–324
- Chang CC, Lin CJ (2011) LIBSVM: a library for support vector machines. *ACM Trans Intell Syst Technol* 2(3):1–27
- Changklom J, Phakawat L, Adichai P (2022) Salinity forecasting on raw water for water supply in the Chao Phraya River. *Water* 14(5):741. <https://doi.org/10.3390/w14050741>
- Chen T, Guestrin C (2016) Xgboost: a scalable tree boosting system. In: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp 785–794
- Cortes C, Vapnik V (1995) Support-vector networks. *Mach Learn* 20:273–297. <https://doi.org/10.1007/BF00994018>
- Dang LTT, Tran AH, Vu TLT, Nguyen TAT (2023) Assessment of the impact of climate change and cultivation conditions on rice yield in Vietnamese Mekong Delta. A case study in Vinh Thanh District, Can Tho city. In: *IOP conference series: earth and environmental science*, p 1170012001.
- Dang TTL, Nguyen KP (2019) Influence of meteorological sensitive parameters on rice yield in Vinh Long Province. *J Hydro Meteorol* 638(1):31–37
- Dong X, Tao T, Liu S, Liu S, Xia Y, Yu Y (2010) Prediction of salt water intrusion using BP-RAGA coupled neural network model. In: *3rd international congress on image and signal processing IEEE, Yantai, China*, pp 4244–4248
- Drucker H, Burges CJC, Kaufman L, Smola AJ, Vapnik V (1997) Support vector regression machines. *Adv Neural Inf Process Syst* 9:155–161
- Dung DT, Man MD, Bui DD, William S, Thang TV (2021) Livelihood vulnerability and adaptability of coastal communities to extreme drought and salinity intrusion in the Vietnamese Mekong Delta. *Int J Disast Risk Reduct* 57:102–183. <https://doi.org/10.1016/j.ijdrr.2021.102183>
- Dung DT, Edward P, Huynh TNT, Nguyen DT, Vo HT, Pham TAN, Can TV, Pham KL, Huu LH, Chau NXQ (2022) Climate change impacts on rice based livelihood vulnerability in the lower Vietnamese Mekong Delta: empirical evidence from Can Tho City and Tra Vinh Province. *Environ Technol Innov*. <https://doi.org/10.1016/j.eti.2022.102834>
- Etemad-Shahidi A, Dorostkar A, Liu W-C (2008) Prediction of salinity intrusion in Danshuei Estuarine system. *Water Policy*. <https://doi.org/10.2166/nh.2008.107>
- Filippi M, Preziosa P, Banwell BL, Barkhof F, Ciccarelli O, De Stefano N, Geurts JJG, Paul F, Reich DS, Toosy AT, Traboulsee A, Wattjes MP, Yousry TA, Gass A, Lubetzki C, Weinshenker BG, Rocca MA (2019) Assessment of lesions on magnetic resonance imaging in multiple sclerosis: practical guidelines. *Brain* 142:1858–1875
- Friedman JH (2001) Greedy function approximation: a gradient boosting machine. *Ann Stat* 29(5):1189–1232
- GSO (General Statistics Office) (2016) Statistical yearbook of Vietnam 2020. Statistical Publishing House, Hanoi
- GSO (General Statistics Office) (2020) Statistical yearbook of Vietnam 2020. Statistical Publishing House, Hanoi
- Guo Q, Huang J, Zhou Z, Wang J (2019) Experiment and numerical simulation of seawater intrusion under the influences of tidal fluctuation and groundwater exploitation in coastal multilayered aquifers. *Geofluids*. <https://doi.org/10.1155/2019/2316271>
- Haddout S, Maslouhi A, Magrane B, Igouzal M (2015) Study of salinity variation in the Sebou River Estuary (Morocco). *Desalin Water Treat* 57:1–12. <https://doi.org/10.1080/19443994.2015.1091993>
- Haddout S, Maslouhi A (2018) One-dimensional hydraulic analysis of the effect of sea level rise on salinity intrusion in the Sebou Estuary. *Morocco Mar Geod* 41:270–288
- He M, Zhong L, Sandhu P, Zhou Y (2020) Emulation of a process-based salinity generator for the Sacramento–San Joaquin Delta of California via deep learning. *Water* 12(8):2088
- Heaton J (2015) AIFH. Volume 3: deep learning and neural networks. Heaton Research, Inc., Chesterfield
- Hochreiter S, Schmidhuber J (1997) LSTM can solve hard long time lag problems. In: *Advances in neural information processing systems*, vol 9, MIT Press, Cambridge
- Hu J, Liu B, Peng S (2019) Forecasting salinity time series using RF and ELM approaches coupled with decomposition techniques. *Stoch Environ Res Risk Assess* 33:1117–1135
- Jaramillo J, Velasquez JD, Franco CJ (2017) Research in financial time series forecasting with SVM: contributions from literature. *IEEE Lat Am Trans* 15(1):145–153
- Jindal H, Oberoi AS, Sandhu IS, Chitkara M (2020) Potential porous mediums for electrochemical hydrogen storage: state of art and comparative study. *Mater Today Proc* 21:1888–1898
- Kashefi AM, Ah J, Shafiey Z (2010) Investigation on salinization of karun caused by tide using modeling. 99–110
- Keilholz P, Disse M, Halik Ü (2015) Effects of land use and climate change on groundwater and ecosystems at the middle reaches of

- the Tarim River using the MIKE SHE integrated hydrological model. *Water*. <https://doi.org/10.3390/w7063040>
- Ketabchi H, Mahmoodzadeh D, Ataie-Ashtiani B, Simmons CT (2016) Sea-level rise impacts on seawater intrusion in coastal aquifers: review and integration. *J Hydrol* 535:235–255
- Khan MA, Shah M, Javed MF, Khan M, Rasheed S, El-Shorbagy M, Malik M (2021) Application of random forest for modelling of surface water salinity. *Ain Shams Eng J*. <https://doi.org/10.1016/j.asej.2021.11.004>
- Kovacevic M, Ivanišević N, Dašić T, Marković L (2018) Application of artificial neural networks for hydrological modelling in karst. *Građevinar* 70(01):1–10
- Le TTD, Ishidaira H, Phung KN (2023) Exploring the utility of the entropy method for classifying household livelihood vulnerability and adaptation strategies to salinity intrusion impact-empirical evidence in coastal communities of the Vietnamese Mekong delta. *Environ Dev* 47:2211–4645
- Li Y, Ma W (2010) Applications of artificial neural networks in financial economics: a survey. In: International symposium on computational intelligence and design, Hangzhou, China, pp 211–214. <https://doi.org/10.1109/ISCID.2010.70>
- Lin Z, Qi J (2017) Hydro-dam—a nature-based solution or an ecological problem: the fate of the Tonlé Sap Lake. *Environ Res* 158:24–32
- Lin K, Lu P, Xu C-Y, Yu X, Lan T, Chen X (2019) Modeling saltwater intrusion using an integrated Bayesian model averaging method in the Pearl River Delta. *J Hydro Inf*. <https://doi.org/10.2166/hydro.2019.073>
- Liu BJ, Yan SL, Chen XH, Lian YQ, Xin YB (2014) Wavelet analysis of the dynamic characteristics of saltwater intrusion—a case study in the Pearl River Estuary of China. *Ocean Coast Manag* 95(4):81–92. <https://doi.org/10.1016/j.ocecoaman.2014.03.027>
- Machado RMA, Ricardo PS (2017) Soil salinity: effect on vegetable crop growth. Management practices to prevent and mitigate soil salinization. *Horticulturae* 3(2):30. <https://doi.org/10.3390/horticulturae3020030>
- Maheswaran R, Khosa R (2012) Comparative study of different wavelets for hydrologic forecasting. *Comput Geosci* 46:284–295. <https://doi.org/10.1016/j.cageo.2011.12.015>
- Maleki M, Banihabib ME, Randhir T (2021) SWAT-S: a SWAT-salinity module for watershed-scale modeling of natural salinity. *Environ Model Softw* 135:104906. <https://doi.org/10.1016/j.envsoft.2020.104906>
- Marcelis LFM, Van Den Boogaard R, Meinen E (2000) Control of crop growth and nutrient supply by the combined use of crop models and plant sensors. *IFAC Proc* 33(19):161–166. [https://doi.org/10.1016/S1474-6670\(17\)40906-2](https://doi.org/10.1016/S1474-6670(17)40906-2)
- Melesse AM, Khosravi K, Tiefenbacher JP, Heddarn S, Kim S, Mosavi A, Pham BT (2020) River water salinity prediction using hybrid machine learning models. *Water* 12(10):29–51. <https://doi.org/10.3390/w12102951>
- Mishra A (2018) A decision support system in healthcare prediction. In: Advanced computational and communication paradigms: proceedings of international conference on ICACCP 2017. <https://doi.org/10.1007/978-981-10-8240-5>
- Mittelstet A, Storm D, Stoecker A (2015) Using SWAT to simulate crop yields and salinity levels in the North Fork River Basin, USA. *Int J Agric Biol Eng* 8:110–124
- Nguyen KD, Ha N, Trang H, Nguyen V (2016) Simulating impacts of sea level rise on salinity intrusion in the Mekong Delta, Vietnam in the period 2015–2100 using MIKE 11. *Naresuan Univ Eng J* 11:21–24. <https://doi.org/10.14456/nuej.2016.4>
- Nguyen QH, Tran DD, Dang KK et al (2020) Land-use dynamics in the Mekong delta: from national policy to livelihood sustainability. *Sustain Dev* 28:448–467. <https://doi.org/10.1002/sd.2036>
- Nicholas F, William PA Jr (2022) The impact of sea-level rise on saltwater intrusion for barrier-island aquifers in North Carolina. *Cont Shelf Res* 244:104789. <https://doi.org/10.1016/j.csr.104789>
- Nicholls R, Cazenave A (2010) Sea-level rise and its impact on coastal zones. *Science* 328:1517–1520. <https://doi.org/10.1126/science.1185782>
- Nicolas G, Georges C, Sébastien P (2023) Predicting sea surface salinity in a tidal estuary with machine learning. *Oceanologia* 65(2):318–332
- Pao-Shan Y, Shien-Tsung C, Chang IF (2006) Support vector regression for real-time flood stage forecasting. *J Hydrol* 328(3–4):704–716. <https://doi.org/10.1016/j.jhydrol.2006.01.021>
- Pereira CS, Lopes I, Abrantes I, Sousa JP, Chelinho S (2019) Salinization effects on coastal ecosystems: a terrestrial model ecosystem approach. *Phil Trans R Soc B* 374:20180251. <https://doi.org/10.1098/rstb.2018.0251>
- Pritchard DW (1955) Estuarine circulation patterns. *Proc Am Soc Civ Eng* 81(717)
- Qiu C, Wan Y (2013) Time series modelling and prediction of salinity in the Caloosahatchee River Estuary. *Water Resour Res* 49(9):5804–5816
- Quang TD, Cao DN, Yi CC, Kumar MP (2014) Modeling the influence of river flow and salinity intrusion in the Mekong River Estuary, Vietnam. *Lowland Technol Int* 16(1):14–25, Released on J-STAGE June 28, 2016, Online ISSN 2187-8870, Print ISSN 1344-9656. https://doi.org/10.14247/lti.16.1_14
- Ralston DK, Geyer WR, Lerczak JA (2008a) Subtidal salinity and velocity in the Hudson River Estuary: observations and modeling. *J Phys Oceanogr* 38:753–770
- Ralston DK, Geyer WR, Lerczak JA (2008b) Subtidal salinity and velocity in the Hudson River estuary: observations and modeling. *J Phys Oceanogr* 28:753–770
- Rath J, Hutton P, Chen L and Roy S (2017) A hybrid empirical-Bayesian artificial neural network model of salinity in the San Francisco Bay-Delta estuary-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>). *Environ Modell Softw* 93:193–208. <https://doi.org/10.1016/j.envsoft.2017.03.022>
- Robinson G, Moutari S, Ahmed AA et al (2018) An advanced calibration method for image analysis in laboratory-scale seawater intrusion problems. *Water Resour Manage* 32:3087–3102. <https://doi.org/10.1007/s11269-018-1977-6>
- Rohmer J, Brisset N (2017) Short-term forecasting of saltwater occurrence at La Comté River (French Guiana) using a kernel-based support vector machine. *Environ Earth Sci* 76(6):246
- Saran SH, Rahman A, Yunus A (2018) Simulation of flow and salinity in Rupsha-Passur river system. In: Proceedings of the 4th international conference on civil engineering for sustainable development (ICCESD 2018), KUET, Khulna, Bangladesh
- Sayidjakhon K, Rashid K, Fadong L, van Andre A, Harm B, Ilhomjon A, Komolitdin S, Nabijon K, Hongguang L, Gang C (2023) Impact assessment of soil salinity on crop production in Uzbekistan and its global significance. *Agric, Ecosyst Environ*. <https://doi.org/10.1016/j.agee.2022.108262>
- Shaha DC, Cho YK, Kim TW (2013) Effects of river discharge and tide driven sea level variation on saltwater intrusion in Sumjin River estuary: an application of finitevolume coastal ocean model. *J Coast Res* 29(2):460–470
- Shahid N, Rappon T, Berta W (2019) Applications of artificial neural networks in health care organizational decision-making: a scoping review. *PLoS ONE* 14(2):e0212356. <https://doi.org/10.1371/journal.pone.0212356>. (PMID:30779785; PMID:PMC6380578)
- Shen R, Huang A, Li B, Guo J (2019) Construction of a drought monitoring model using deep learning based on multi-source remote sensing data. *Int J Appl Earth Obs Geoinf* 79:48–57

- Singh A, Kumar S, Akula S, Lawrence DM, Lombardozi DL (2020) Plant growth nullifies the effect of increased water-use efficiency on streamflow under elevated CO₂ in the South-eastern United States. *Geophys Res Lett* 47:e2019GL086940. <https://doi.org/10.1029/2019GL086940>
- Soc Trang Statistical Yearbook (2016) Statistical yearbook of Soc Trang 2015. Statistical Publishing House, Hanoi
- Soc Trang Statistical Yearbook (2017) Statistical yearbook of Soc Trang 2016. Statistical Publishing House, Hanoi
- Soc Trang Statistical Yearbook (2018) Statistical yearbook of Soc Trang 2017. Statistical Publishing House, Hanoi
- Soc Trang Statistical Yearbook (2019) Statistical yearbook of Soc Trang 2018. Statistical Publishing House, Hanoi
- Soc Trang Statistical Yearbook (2020) Statistical yearbook of Soc Trang 2019. Statistical Publishing House, Hanoi
- Soc Trang Statistical Yearbook (2021) Statistical yearbook of Soc Trang 2020. Statistical Publishing House, Hanoi
- Somboonsuke B, Phitthayaphinant P, Sdoodee S, Kongmanee C (2018) Farmers' perceptions of impacts of climate variability on agriculture and adaptation strategies in Songkhla Lake basin. *Kasetsart J Soc Sci* 39(2):277–283. <https://doi.org/10.1016/j.kjss.2018.05.006>
- Sun H, Koch M (2001) Case study: analysis and forecasting of salinity in Apalachicola Bay, Florida, using Box-Jenkins ARIMA models. *J Hydraul Eng* 127(9):718–727
- Taylor KE (2001) Summarizing multiple aspects of model performance in a single diagram. *J Geophys Res* 106:7183–7192
- Tessler Z, Vörösmarty C, Grossberg M, Gladkova I, Aizenman H, Syvitski J, Foufoula-Georgiou E (2015) Profiling risk and sustainability in coastal deltas of the world. *Science* 349:638–643. <https://doi.org/10.1126/science.aab3574>
- Thanh TN, Van Huynh H, Vo Minh H, Tri VPD (2023) Salinity intrusion trends under the impacts of upstream discharge and sea level rise along the Co Chien river and Hau river in the Vietnamese Mekong Delta. *Climate* 11(3):66. <https://doi.org/10.3390/cli11030066>
- Tho N, Vromant N, Hung NT, Hens L (2008) Soil salinity and sodicity in a shrimp farming coastal area of the Mekong Delta, Vietnam. *Environ Geol* 54:1739–1746
- Tran DA, Tsujimura M, Ha NT, Nguyen VT, Binh DV, Dang TD, Doan Q-V, Bui DT, Anh Ngoc T, Phu LV, Thuc PTB, Pham TD (2021) Evaluating the predictive power of different machine learning algorithms for groundwater salinity prediction of multi-layer coastal aquifers in the Mekong Delta, Vietnam. *Ecol Indic* 127:1–14. <https://doi.org/10.1016/j.ecolind.2021.107790>
- Tran TT, Pham NH, Pham QB et al (2022) Performances of different machine learning algorithms for predicting saltwater intrusion in the Vietnamese Mekong Delta using limited input data: a study from Ham Luong river. *Water Resour* 49:391–401. <https://doi.org/10.1134/S0097807822030198>
- Tuan LA (2010) Impacts of climate change and sea level rise on the biodiversity and human migration trend of Ca Mau Peninsula, the Mekong River Delta. *Ca Mau*
- Uncles RJ, Stephens JA (2011) The effects of wind, runoff and tides on salinity in a strongly tidal sub-estuary. *Estuar Coasts* 34(4):758–774
- United Nations (2022) The united nations world water development report: groundwater: making the invisible visible. UNESCO, Paris
- Van Houdt G, Mosquera C, Nápoles G (2020) A review on the long short-term memory model. *Artif Intell Rev* 53:5929. <https://doi.org/10.1007/s10462-020-09838->
- Wang J (2013) Partial correlation coefficient. In: Dubitzky W, Wolkenhauer O, Cho KH, Yokota H (eds) *Encyclopedia of systems biology*. Springer, New York. https://doi.org/10.1007/978-1-4419-9863-7_373
- Wawargita PW, Rahmat AP (2016) Adaptation of social-economic livelihoods in coastal community: the case of Mangunharjo sub-district, Semarang city. *Proc Soc Behav Sci* 227:477–484. <https://doi.org/10.1016/j.sbspro.2016.06.103>
- Willmott CJ (1982) Some comments on the evaluation of model performance. *Bull Am Meteor Soc* 63(11):1309–1313
- Willmott CJ, Matsuura K (2005) Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Clim Res* 30: 79–82. <https://doi.org/10.3354/cr030079>
- Xu C, McDowell NG, Fisher RA, Wei L, Sevanto S, Christoffersen BO, Weng E, Middleton RS (2019a) Increasing impacts of extreme droughts on vegetation productivity under climate change. *Nat Clim Change* 9(12):948–953. <https://doi.org/10.1038/s41558-019-0630-6>
- Xu Y, Hoitink AJF, Kästner K, Zhang W (2019b) Analytical model captures intratidal variation in salinity in a convergent, well-mixed estuary. *Hydrol Earth Syst Sci* 23:4309–4322. <https://doi.org/10.5194/hess-23-4309-2019>
- Xu J, Feng Z, Tang J, Liu S, Ding Z, Lyu J, Yao Q, Yang B (2022) Improved random forest for the automatic identification of *Spodoptera frugiperda* larval instar stages. *Agriculture* 12(11):1919. <https://doi.org/10.3390/agriculture12111919>
- Yang X, Guo S, Deng X, Wang W, Xu D (2022) Study on livelihood vulnerability and adaptation strategies of farmers in areas threatened by different disaster types under climate change. *Agriculture* 11:1088. <https://doi.org/10.3390/agriculture1111>
- Yoon H, Kim Y, Ha K, Lee SH, Kim GP (2017) Comparative evaluation of ANN- and SVM-time series models for predicting freshwater-saltwater interface fluctuations. *Water* 9(5):323
- Zar JH (1999) *Biostatistical analysis*. Prentice Hall, Upper Saddle River
- Zewdie GK, Lary DJ, Liu X, Wu D, Levetin E (2019) Estimating the daily pollen concentration in the atmosphere using machine learning and NEXRAD weather radar data. *Environ Monit Assess* 191:1–9
- Zhou F, Liu B, Duan K (2020) Coupling wavelet transform and artificial neural network for forecasting estuarine salinity. *J Hydrol* 558:125–127. <https://doi.org/10.1016/j.jhydrol.2020.125127>
- Zhou J, Deitch M, Grunwald S, Sreaton E, Olabarrieta M (2021) Effect of Mississippi River discharge and local hydrological variables on salinity of nearby estuaries using a machine learning algorithm. *Estuar Coast Shelf Sci* 263:107628. <https://doi.org/10.1016/j.ecss.2021.107628>
- Zhou Z, Li X, Wu H (2016) Wind power prediction based on random forests. In: 2016 4th international conference on electrical & electronics engineering and computer science (ICEECS 2016). <https://doi.org/10.2991/iceeecs-16.2016.73>
- Zou M, Jiang W-G, Qin Q-H, Liu Y-C, Li M-L (2022) Optimized XGBoost model with small dataset for predicting relative density of Ti-6Al-4V parts manufactured by selective laser melting. *Materials* 15:5298. <https://doi.org/10.3390/ma15155298>

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